

# Discrete Random Expected Utility and Self-Selection

Igor Kopylov\*

Kevin Peng<sup>†‡</sup>

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## WORKING DRAFT JOB MARKET PAPER

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**SELECT SLIDES FOR ASSOCIATED EXPERIMENTAL PROJECT  
INCLUDED IN APPENDIX B.** (Peng, 2026)

### Abstract

We model stochastic choice rules via finitely many types  $\theta$  that maximize distinct expected utility functions and use endogenous tie-breaking rules. First, we characterize *discrete random expected utility* (DREU) where the likelihood  $\mu(\theta)$  of each relevant type  $\theta$  is preserved across all menus  $A$ . This model is a discrete version for the *random expected utility* of Gul and Pesendorfer (2006), but our axioms, identification, and tie-breaking procedures are novel. More generally, we propose *discrete-map expected utility* (DMEU) where the likelihoods  $\mu_A(\theta)$  can vary continuously with the menu  $A$ . The map  $\mu$  is identified uniquely in our model. DMEU captures various kinds of *context dependence* with both behavioral and normative motivations. Our main application of DMEU is a model of *self-selection*, where types can increase their participation rates across distinct menus when their best choices are improved. The standard *monotonicity* principle for stochastic choices delivers a *self-selective* property for the map  $\mu$ . We argue that self-selection in menus of lotteries can be consistent with the Block–Marschak random utility model over menus of deterministic outcomes, but the identification of types in the latter approach becomes misspecified. Finally, we discuss applications of DREU and DMEU to random *risk aversion* and random *Cobb–Douglas utility*.

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\*Department of Economics, University of California, Irvine. E-mail: ikopylov@uci.edu.

<sup>†</sup>Department of Economics, University of California, Irvine. E-mail: kuangyup@uci.edu

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# 1 Introduction

Observed choices in consumption menus can be naturally *stochastic* when they are produced by distinct rational (i.e., utility maximizing) agents. It is common for data to be presented as an aggregation of choices from diverse agents. Such data is used in a wide range of empirical work with examples of preferred modes of transportation (McFadden, 2001) and choice of fishing sites (Train, 1998). This heterogeneity may also result from more behavioral motivations where a single individual can exhibit distinct random responses in identical choice problems (see Agranov and Ortoleva (2017)).

The classic random utility model (RUM) by Block and Marschak (1960) (henceforth, BM) aims to represent stochastic choice data by a stable probability distribution  $\mu$  across *types* that include all possible utility functions. In the general case where types are unrestricted, Falmagne (1978) characterizes RUM via non-negativity of Block and Marschak's polynomials. Gul and Pesendorfer (2006) (henceforth, GP) refine this model to *random expected utility* (REU) where all types should maximize expected utility functions rather than arbitrary ones. Other refinements of RUM have restricted types by single-crossing conditions (Apestegui, Ballester, and Lu, 2017), quasi-linearity (McFadden, 1973; Williams, 1977; Daly and Zachery, 1979; Yang and Kopylov, 2023), and other assumptions. Behavioral patterns, such as ambiguity aversion (Lu, 2021), have also been discussed within the context of random utility. All of these refinements preserve the assumption that  $\mu$  is invariant of the consumption menu.

In this paper, we propose a novel version of random utility that has

- a discrete (i.e., finite) type space  $\Theta$  where all types  $\theta \in \Theta$  combine expected utility maximization with endogenous tie-breaking rules;
- a distribution of types  $\mu_A$  that can depend on the menu  $A$  and capture context dependence and self-selection;
- a unique identification for both  $\Theta$  and  $\mu_A$ .

The case where the distribution  $\mu(\theta)$  is invariant of  $A$  is a discrete version of the GP's REU, but our axioms and identifications are more parsimonious than theirs and rely more on algebraic rather than analytical techniques. The novel identification approach allows the consumption domain  $X$  to be more flexible than in GP, where  $X$  is a finite-dimensional simplex of lotteries. In our framework,  $X$  is convex, but need not be closed, or bounded, or finite dimensional. For example,  $X$  may consist of monetary gambles with payoffs on the real line, or  $X$  can be the non-negative orthant of consumption bundles.

More substantially, we depart from REU by allowing the distribution  $\mu_A$  to vary continuously with the menu  $A$ . In general, it is only required that the support  $\Theta = \{\theta : \mu_A(\theta) > 0\}$  be finite and invariant of  $A$ . We use the term *discrete maps* to refer to such functions  $\mu$  that map menus  $A$  into distributions  $\mu_A$  over types. Our most general representation takes the form

$$\rho(x, A) = \mu_A\{\theta \in \Theta : x \text{ is the best element for } \theta \text{ in } A\} \quad (1)$$

where  $\mu_A(\theta)$  is the probability of type  $\theta$  making a choice in the menu  $A$ .

Each type  $\theta$  is assumed to maximize some expected utility function  $u_\theta$  in each menu  $A$ . However, in order for representation (1) to be well-defined, each type  $\theta \in \Theta$  must also incorporate some endogenous tie-breaking rule that selects a unique maximizer for  $u_\theta$  in  $A$ . Formally, each  $\theta$  is associated with a total, complete, transitive binary relation that satisfies von Neumann-Morgenstern's Independence axiom as well. Such total types have been used in finite-dimensional contexts by Fishburn (1982), Myerson (1986), Blume, Branderburger, and Dekel (1991) and others. We argue that there is a rich family of total types even if  $X$  is not finite dimensional. Thus we provide a novel way to accommodate tie-breaking in random expected utility. Without totality, Piermont (2022) relaxes GP's model to represent only those choices that are strict maximizers for corresponding types.

We show several characterization results for stochastic choice rules. First, Theorem 2 below characterizes representation (1) where  $\mu_A = \mu$  is menu-invariant and relates this result to GP's model. Our main result (Theorem 3 below) characterizes representation (1) and provides a convenient platform for various refinements that impose more structure on the menu-contingent distribution  $\mu_A$  and/or types  $\theta$ .

DMEU weakens an axiom from DREU and requires the discussion of elements being chosen with positive probability. In empirical settings, however, observed choice frequencies may be affected by noise, making it inappropriate to treat any strictly positive empirical frequency as evidence of positive choice probability. Since even an alternative that is never chosen in the data may have a small but nonzero underlying probability, an approximate criterion is required.

A useful benchmark is provided by the Rule of Three, which states that if an event is not observed in  $N$  independent trials, then with 95% confidence its true rate of occurrence is at most  $\frac{3}{N}$  (Jovanovic and Levy, 1997). Accordingly, in a population of size  $N$ , this bound yields a conservative threshold: alternatives with true choice probabilities exceeding  $\frac{3}{N}$  would be expected to be chosen by at least one agent with high probability. We therefore treat  $\frac{3}{N}$  as a reasonable cutoff for considering an element as effectively chosen.

Our main application of DMEU assumes the standard monotonicity principle that asserts

$$\rho(x, A) \geq \rho(x, A \cup B)$$

for all menus  $A, B \in \mathcal{M}$  and elements  $x \in A$ . Theorem 4 establishes that monotonicity in DMEU is equivalent to the *self-selective* property of the map  $\mu$ :

$$\mu_A(\theta) \geq \mu_{A \cup B}(\theta)$$

for all types  $\theta \in \Theta$  that have the same maximal element in  $A$  and in  $A \cup B$ . This property reflects a notion of self-selection where types can increase their participation in the menu  $A \cup B$ , but only if they find better choices in the larger menu. For example, it should be reasonable for gamblers to show up in higher numbers to bookmakers that offer them more favorable odds on some sporting event. What's more, the expansion of bet types with the rise of online gambling can incentivize new customers to enter the market (Hing et al., 2022). Self-selection can also be observed in the dining industry. Garnett et al. (2019) found that doubling the options of vegetarian meals increased sales of such meals by around 15%.

For example, assume that menus consist of lotteries on a finite set  $Q$  of deterministic outcomes. Then the self-selective property is implied if for all menus  $A$  and types  $\theta \in \Theta$ ,

$$\mu_A(\theta) = \frac{\max_{x \in A} u_\theta(x)}{\sum_{\psi \in \Theta} \max_{x \in A} u_\psi(x)}, \quad (2)$$

where the utility function  $u_\theta$  is such that the best choice for  $\theta$  in  $A$  must maximize  $u_\theta$  in  $A$ . Here the maximal value  $\max_{x \in A} u_\theta(x)$  can be interpreted as a total number of agents of type  $\theta$  that find it reasonable to participate in the menu  $A$ . Moreover, we argue that the Luce-like specification (2) implies the random utility model for the restriction of  $\rho$  to menus  $A \subset Q$  that consist of deterministic outcomes. However, the identification of types in the latter approach becomes misspecified. In other words, the random utility model does not hold for the true set of types  $\Theta$ , but instead distorts  $\Theta$  by phantom types.

Empirical evidence highlights that the standard RUM can fail to disentangle underlying decision mechanisms when self-selection and limited consideration are important features of choice behavior. For instance, Heiss et al. (2021) analyze administrative data from Medicare Part D to show that observed consumer inertia—where only about 10% of beneficiaries switch plans annually—can arise from either inattention to plan choice or high switching costs, and that separating these mechanisms is crucial for credible counterfactuals on policy interventions such as defaults or decision aids. Similarly, Abaluck and Adams-Prassl (2021) demonstrate through both a controlled lab experiment and application to Medicare Part D

choices that consideration set formation and preference heterogeneity can be identified from choice data only under additional structure, since different combinations of consideration probabilities and utilities can generate the same observed choice probabilities. These studies illustrate that without richer modeling of participation, attention, or self-selection, conventional RUM specifications can be underidentified: multiple structural interpretations are consistent with the same choice patterns, leading to potentially divergent policy conclusions.

More broadly, there are some behavioral patterns, such as reasoned-based heuristics in Shafir, Simonson, and Tversky (1993), where Monotonicity does not hold and hence, self-selection is no longer a plausible explanation. We use *context dependence* as a blanket term to refer to all such patterns (see a literature review by Rabin (1998)). An example of such patterns is extremeness aversion. Sharpe et al. (2008) observed that the removal of the largest drink size (44 oz) from the menu led consumers to shift their purchases from the 32-oz option—now the largest available size—to the 21-oz size.

Finally, we discuss applications of DMEU and DREU to random risk aversion, beliefs, and Cobb-Douglas utility functions. In such refinements, the type space  $\Theta$  can be restricted to the required domain by extra conditions on  $\rho$ . For example, agents make more risk neutral choices when lotteries are presented in a binary format (i.e., only two states of the world have positive probabilities) instead of lotteries that are more complex or compound lotteries (Harrison et al., 2013). Changes in beliefs, which shall be discussed in detail in a separate project, can arise in the form of partition dependence. Ahn and Ergin (2010) observed that agents, when choosing deductible plans, place a higher weight on surgeries when they are individually listed than when they are bundled under the umbrella term “surgery”. In such situations, the consumption domain  $X$  can be taken to consist of lotteries or Anscombe-Aumann (1963) uncertain prospects.

## 2 Preliminaries

Let  $X = \{x, y, \dots\}$  be a non-singleton consumption domain. Assume that  $X$  is a convex subset of a normed linear space  $L$ .

For example, take any set  $Q$  of deterministic prizes (e.g. monetary ones). Let  $\Delta^Q$  be the set of all *lotteries*: functions  $x : Q \rightarrow [0, 1]$  such that  $x(q) > 0$  is positive for finitely many  $q \in Q$ , and  $\sum_{q \in Q} x(q) = 1$ . Note that  $X = \Delta^Q$  is a convex subset of the linear space  $L = \mathbb{R}^Q$  endowed with the uniform norm. In the random expected utility model, GP take  $X = \Delta^n$  for some finite  $Q = \{1, \dots, n\}$ . In our framework,  $X$  need not be finite-dimensional, closed, or bounded, and elements in  $X$  may have interpretations other than lotteries.

A binary relation  $\theta$  on  $X$  is called a *type* if it is complete, transitive, and satisfies *Inde-*

*pendence*: for all  $x, y, z \in X$  and  $\gamma \in (0, 1]$ ,

$$x\theta y \Leftrightarrow [\gamma x + (1 - \gamma)z]\theta[\gamma y + (1 - \gamma)z].$$

Let  $\theta^*$  be the asymmetric part of the type  $\theta$ . Thus  $\theta$  and  $\theta^*$  denote a special case of binary relations that are often written as  $\succeq$  and  $\succ$  in decision-theoretic literature.

A type  $\theta$  is called *null* if  $x\theta y$  for all  $x, y \in X$ . Otherwise,  $\theta$  is called *non-null*, and  $x\theta^*y$  is strict for some  $x, y \in X$ .

Let  $\mathcal{M} = \{A, B, \dots\}$  be the set of all *menus*—non-empty finite subsets of  $X$ . Singleton menus  $\{x\}$  are written as  $x$ .

For any type  $\theta$  and menu  $A \in \mathcal{M}$ , say that  $x \in A$  is *best* for  $\theta$  in  $A$  if  $x\theta y$  for all  $y \in A$ . Such  $x$  exists because  $\theta$  is complete and transitive, and  $A$  is finite.

Let  $\mathcal{T}$  be the set of all *total* types such that for all  $x, y \in X$ ,

$$x\theta y\theta x \Rightarrow x = y.$$

This property implies that for any menu  $A \in \mathcal{M}$  and total type  $\theta \in \mathcal{T}$ , there is a unique element in  $A$  that is best for  $\theta$  in  $A$ .

Take any menu  $A \in \mathcal{M}$ . For any feasible element  $x \in A$ , let

$$\mathcal{T}(x, A) = \{\theta \in \mathcal{T} : x \text{ is best for } \theta \text{ in } A\}.$$

Then  $\mathcal{T}$  is *partitioned* into a disjoint union

$$\mathcal{T} = \bigcup_{x \in A} \mathcal{T}(x, A). \quad (3)$$

because for each  $\theta \in \mathcal{T}$ , there is one and only one  $x \in A$  such that  $\theta \in \mathcal{T}(x, A)$ .

For any  $A, B \in \mathcal{M}$  and  $\gamma \in [0, 1]$ , define a mixture

$$\gamma A + (1 - \gamma)B = \{\gamma x + (1 - \gamma)y : x \in A, y \in B\}.$$

If  $\gamma \in (0, 1]$ , then for any  $x \in A$ , the set  $\mathcal{T}(x, A)$  is partitioned by

$$\mathcal{T}(x, A) = \bigcup_{z \in [\gamma x + (1 - \gamma)B]} \mathcal{T}(z, \gamma A + (1 - \gamma)B). \quad (4)$$

Indeed, if  $\gamma \in (0, 1]$ ,  $x \in A$ ,  $y \in B$ ,  $\theta \in \mathcal{T}$ , and  $b$  is best for  $\theta$  in  $B$ , then

- $x$  is best for  $\theta$  in  $A$  implies that  $z = \gamma x + (1 - \gamma)b$  is best for  $\theta$  in  $\gamma A + (1 - \gamma)B$ ;

- otherwise, no  $z = \gamma x + (1 - \gamma)y$  is best for  $\theta$  in  $\gamma A + (1 - \gamma)B$ .

Thus the disjoint union (4) holds.

## 2.1 Mixture Linear Utility Maximization

A function  $u : X \rightarrow \mathbb{R}$  is called *mixture linear* if for all  $x, y \in X$  and  $\gamma \in [0, 1]$ ,

$$u(\gamma x + (1 - \gamma)y) = \gamma u(x) + (1 - \gamma)u(y).$$

If  $X = \Delta^Q$ , then mixture linearity implies the *expected utility* form:

$$u(x) = \sum_{q \in Q} x(q)u(q) \quad \text{for all } x \in X.$$

In general, expected utility is not the only possible motivation for mixture linearity. For example, it holds for Cobb-Douglas utility functions over the positive orthant of consumption bundles in the logarithmic scale.

Let  $\mathcal{U}$  be the set of all non-constant mixture linear functions  $u : X \rightarrow \mathbb{R}$ . For any  $u \in \mathcal{U}$ , let  $\theta_u$  be the binary relation that is represented by  $u$ : for all  $x, y \in X$ ,

$$x\theta_u y \quad \Leftrightarrow \quad u(x) \geq u(y).$$

A type  $\theta$  is called *mixture continuous* if for all  $x, y, z \in X$ ,

$$\{\gamma \in [0, 1] : z\theta[\gamma x + (1 - \gamma)y]\} \quad \text{and} \quad \{\gamma \in [0, 1] : [\gamma x + (1 - \gamma)y]\theta z\}$$

are closed sets. Otherwise,  $\theta$  is *discontinuous*.

Let  $\mathcal{C}$  be the class of all types that are mixture continuous and non-null. Herstein and Milnor's (1953) *mixture space theorem* asserts that

$$\theta \in \mathcal{C} \quad \Leftrightarrow \quad \theta = \theta_u \quad \text{for some } u \in \mathcal{U}.$$

Moreover, such  $u$  is unique up to a positive linear transformation: if  $\theta_u = \theta_v$  for some  $u, v \in \mathcal{U}$ , then  $v = \alpha u + \beta$  for some  $\alpha > 0$  and  $\beta \in \mathbb{R}$ .

However, total types can be mixture continuous only if  $X$  is a convex subset of a line  $\{x + \alpha d : \alpha \in \mathbb{R}\}$  for some  $x, d \in L$  such that  $d \neq 0$ . Call such  $X$  *collinear*. Then the functions  $u_+(x + \alpha d) = \alpha$  and  $u_-(x + \alpha d) = -\alpha$  represent the only two non-null types  $\theta_{u_+}$  and  $\theta_{u_-}$ . Both are total and mixture continuous.

In the generic *non-collinear* case,  $X$  is not a subset of any line. Then all total types are discontinuous, that is,  $\mathcal{T} \cap \mathcal{C} = \emptyset$ . To show this, use non-collinearity to find  $x, y, z \in X$  such that  $x - y$  and  $y - z$  are linearly independent. Let  $\theta \in \mathcal{C}$ . Then  $\theta = \theta_u$  for some  $u \in \mathcal{U}$ . Without loss in generality,  $u(x) \geq u(y) \geq u(z)$ . Then either  $u(x) = u(z)$ , or  $u(y) = u(\gamma x + (1 - \gamma)z)$  where  $\gamma = \frac{u(y) - u(z)}{u(x) - u(z)}$ . As  $x \neq z$  and  $y \neq \gamma x + (1 - \gamma)z$ , then  $\theta$  is not total in either case.

In a weaker form, the mixture space theorem can be extended to discontinuous (e.g. total) types. Say that a function  $u : X \rightarrow \mathbb{R}$  is *maximized* by a type  $\theta$  if for all  $x, y \in Z$ ,

$$x\theta y \quad \Rightarrow \quad u(x) \geq u(y). \quad (5)$$

Equivalently,  $u$  is maximized by  $\theta$  if for all  $A \in \mathcal{M}$  and  $x \in A$ ,

$$x \text{ is best for } \theta \text{ in } A \quad \Rightarrow \quad x \text{ is a maximizer for } u \text{ in } A.$$

The one-way representation (5) with arbitrary utility functions is proposed by Richter (1966) and Peleg (1970). As we argue next, if  $u$  is required to be mixture linear in representation (5), then  $u$  can be identified just as sharply as in the standard expected utility model.

A type  $\theta$  is called *stable* if there exist  $x^*, y^* \in X$  such that for any  $z_x, z_y \in X$ , there exists  $\gamma \in (0, 1]$  such that

$$[\gamma z_x + (1 - \gamma)x^*]\theta^*[\gamma z_y + (1 - \gamma)y^*]. \quad (6)$$

The strict comparison  $x^*\theta^*y^*$  is implied here because if  $y^*\theta x^*$ , then (6) fails for  $z_x = x^*$  and  $z_y = y^*$ . Thus a stable type  $\theta$  should allow some strict comparison  $x^*\theta^*y^*$  that is unperturbed by mixtures  $\gamma z_x + (1 - \gamma)x^*$  and  $\gamma z_y + (1 - \gamma)y^*$ , where the positive weight  $\gamma$  can be adapted to  $z_x$  and  $z_y$ . As  $\gamma$  can be arbitrarily small, then stability can be viewed as a local Archimedean property.

**Theorem 1.** *A type  $\theta$  is stable if and only if  $\theta$  maximizes some  $u \in \mathcal{U}$ . Such  $u$  is unique up to a positive linear transformation.*

*Moreover, if  $X$  is finite dimensional, then all non-null types are stable.*

All proofs are in the appendix. For any stable type  $\theta$ , Theorem 1 asserts a mixture linear utility maximization (5) with the same uniqueness claim as in Herstein–Milnor’s theorem. If  $X$  is finite dimensional, then any non-null type is stable and hence, maximizes some  $u \in \mathcal{U}$ . In general, a non-null type  $\theta$  can violate stability. Such  $\theta$  can maximize a mixture linear function  $u$  only in the trivial case when  $u$  is constant and hence, does not belong to  $\mathcal{U}$ .

The utility maximization (5) can be further refined via tie-breaking rules in Theorem 8 below. We also use that result to construct a rich family of stable total types on any convex

domain  $X$ .

### 3 Main Representation Results

Let  $\Omega$  be the set of all pairs  $(x, A)$  such that  $A \in \mathcal{M}$  and  $x \in A$ , that is,  $x$  is a feasible element in a menu  $A$ . Such pairs are called *trials*.

A function  $\rho : \Omega \rightarrow [0, 1]$  is called a *stochastic choice rule* (scr) if

$$\sum_{x \in A} \rho(x, A) = 1 \quad \text{for all } A \in \mathcal{M}. \quad (7)$$

Here, the probability  $\rho(x, A)$  of any trial  $(x, A) \in \Omega$  can be interpreted as the likelihood of  $x$  being chosen when the menu  $A$  is feasible.

A function  $\mu : 2^{\mathcal{T}} \rightarrow \mathbb{R}_+$  is called a *discrete distribution* if

- (i) the set  $\Theta_\mu = \{\theta \in \mathcal{T} : \mu(\theta) > 0\}$  is finite,
- (ii) for all sets  $\Psi \subset \mathcal{T}$ ,  $\mu(\Psi) = \sum_{\theta \in \Psi} \mu(\theta)$ ,
- (iii)  $\mu(\mathcal{T}) = \mu(\Theta_\mu) = 1$ .

By definition,  $\mu$  is a countably additive<sup>1</sup> probability measure. The set  $\Theta_\mu$  is called the *support* of  $\mu$ .

Let  $\mathcal{D}$  be the set of all discrete distributions. Say that a stochastic choice rule  $\rho$  is *represented* by a discrete distribution  $\mu \in \mathcal{D}$  if for any trial  $(x, A) \in \Omega$ ,

$$\rho(x, A) = \mu(\mathcal{T}(x, A)) \quad (8)$$

is the likelihood that  $\mu$  assigns to all total types for which  $x$  is best in  $A$ . If  $X$  is a finite dimensional simplex of lotteries, then by Theorem 1, any total type  $\theta \in \mathcal{T}$  maximizes some expected utility function  $u \in \mathcal{U}$ . Thus we refer to representation (8) as *discrete random expected utility* (DREU).

DREU implies that for any  $(x, A) \in \Omega$ ,

$$\rho(x, A) > 0 \quad \Leftrightarrow \quad x \text{ is best in } A \text{ for some } \theta \in \Theta_\mu. \quad (9)$$

For any menu  $A$ , let the *capacity* of  $\rho$  in  $A$

$$\kappa_\rho(A) = |\{x \in A : \rho(x, A) > 0\}|$$

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<sup>1</sup>For discrete distributions, countable and finite additivity are equivalent.

be the number of all elements that are chosen with a positive probability in  $A$ . As  $\Theta_\mu$  is finite, then by (9),  $\kappa_\rho(A) \leq |\Theta_\mu|$ .

**Axiom 1** (Discreteness). *There is  $n \in \mathbb{N}$  such that  $\kappa_\rho(A) \leq n$  for all  $A \in \mathcal{M}$ .*

Discreteness cannot be rejected by finitely many observations and hence, plays a technical role akin to continuity conditions in GP's REU.

To get empirically testable implications of DREU, take any  $(x, A) \in \Omega$ ,  $B \in \mathcal{M}$ , and  $\gamma \in (0, 1]$ . Let

$$\rho(x, A, \gamma, B) = \sum_{z \in [\gamma x + (1-\gamma)B]} \rho(z, \gamma A + (1-\gamma)B)$$

be the combined probability of all choices in the menu  $\gamma A + (1-\gamma)B$  that belong to its submenu  $\gamma x + (1-\gamma)B$ .

**Axiom 2** (Additivity).  $\rho(x, A) = \rho(x, A, \gamma, B)$ .

This equality follows from (8) where the discrete distribution  $\mu$  is applied to both sides of the partition (4). Additivity implies

**Axiom 3** (Monotonicity).  $\rho(x, A) \geq \rho(x, A \cup B)$ .

Indeed,  $\rho(x, A) = \rho(x, A, \frac{1}{2}, A \cup B) \geq \rho(x, A \cup B, \frac{1}{2}, A) = \rho(x, A \cup B)$ , where the middle inequality follows from  $\frac{1}{2}x + \frac{1}{2}(A \cup B) \supset \frac{1}{2}x + \frac{1}{2}A$ .

Unlike Additivity, Monotonicity does not use mixtures of outcomes or menus, and has a more straightforward interpretation. Monotonicity holds in BM's general random utility model and all of its refinements that take the distribution  $\mu$  to be invariant across menus  $A$ . It holds in DREU as well, where it is subsumed by Additivity.

**Theorem 2.** *A stochastic choice rule  $\rho$  satisfies Discreteness and Additivity if and only if  $\rho$  is represented by some discrete distribution  $\mu \in \mathcal{D}$ . Such  $\mu$  is unique.*

The proof is self-contained and constructive. Its sketch is available after the more general Theorem 3 below. In Section 4, we compare DREU and Theorem 2 with GP's REU and their representation results.

To illustrate the role of total types in Theorem 2, consider another definition. For any  $\theta \in \mathcal{T}$  and  $(x, A) \in \Omega$ , let

$$\rho_\theta(x, A) = \begin{cases} 1 & \text{if } x \text{ is best for } \theta \text{ in } A \\ 0 & \text{otherwise.} \end{cases}$$

Obviously,  $\rho_\theta$  is a stochastic choice rule, and representation (8) asserts that

$$\rho = \sum_{\theta \in \Theta_\mu} \mu(\theta) \rho_\theta$$

is a convex combination of  $\rho_\theta$  across  $\theta \in \Theta_\mu$ . By contrast, there is no convenient way to specify  $\rho_\theta$  when  $\theta$  is not total. For example, the uniform tie-breaking rule violates both Discreteness and Additivity. Indeed, take any type  $\theta$  such that  $x \theta y \theta x$  for some distinct  $x, y \in X$ . Let  $B = \{x, y\}$ ,  $z = \frac{1}{2}x + \frac{1}{2}y$ , and  $C = \frac{1}{2}B + \frac{1}{2}B = \{x, y, z\}$ . Then  $x$  and  $y$  are best for  $\theta$  in  $B$ , and by Independence,  $x$ ,  $y$ , and  $z$  are all best for  $\theta$  in  $C$ . The uniform tie-breaking implies that

$$\begin{aligned} \rho_\theta(x, B) &= \rho_\theta(y, B) = \frac{1}{2}, \\ \rho_\theta(x, C) &= \rho_\theta(y, C) = \rho_\theta(z, C) = \frac{1}{3}, \end{aligned}$$

which violates Additivity because  $\rho(x, B) = \frac{1}{2} \neq \frac{2}{3} = \rho(x, B, \frac{1}{2}, B)$ . Next, for any  $n \in \mathbb{N}$ , consider a menu  $A = \{a_1, \dots, a_n\}$  where  $a_i = \frac{i}{n}x + \frac{n-i}{n}y$ . By Independence, all  $a_i$  are best for  $\theta$  in  $A$ . The uniform tie-breaking implies that  $\rho(a_i, A) = \frac{1}{n}$  for each  $i$ . As  $n$  is arbitrary, then Discreteness fails.

### 3.1 Discrete-Map Expected Utility

DREU and its Additivity axiom can be problematic if the distribution of types in  $\Theta_\mu$  is context dependent. For example, adding extra options to a menu  $A$  can increase the probability of choosing  $x \in A$  due to reason-based heuristics (e.g., Shafir, Simonson, and Tversky (1993)) and other behavioral reasons. Thus Monotonicity and a fortiori, Additivity can be violated.

To accommodate context dependence, discrete distributions can be assigned contingently on the feasible menu  $A$  via some function  $\mu : \mathcal{M} \rightarrow \mathcal{D}$ . It is convenient to write the distribution that  $\mu$  assigns to  $A$  as  $\mu_A$  rather than  $\mu(A)$ .

To impose continuity on  $\mu$ , metrize  $\mathcal{D}$  and  $\mathcal{M}$  by the uniform and Hausdorff metrics respectively such that for all  $\mu, \pi \in \mathcal{D}$  and  $A, B \in \mathcal{M}$ ,

$$\begin{aligned} d_u(\mu, \pi) &= \max_{\Psi \subset \mathcal{T}} |\mu(\Psi) - \pi(\Psi)| \\ d_h(A, B) &= \max \left\{ \max_{x \in A} \min_{y \in B} \|x - y\|, \max_{y \in B} \min_{x \in A} \|x - y\| \right\}. \end{aligned}$$

A continuous function  $\mu : \mathcal{M} \rightarrow \mathcal{D}$  is called a *discrete map* if for all  $A, B \in \mathcal{M}$ ,  $\mu_A$  and

$\mu_B$  share the same support

$$\Theta_\mu = \{\theta \in \mathcal{T} : \mu_A(\theta) > 0\} = \{\theta \in \mathcal{T} : \mu_B(\theta) > 0\}.$$

Let  $\mathcal{D}(\mathcal{M})$  be the set of all discrete maps. Identify any discrete distribution  $\mu \in \mathcal{D}$  with the constant discrete map  $\mu \in \mathcal{D}(\mathcal{M})$ .

For any discrete map  $\mu \in \mathcal{D}(\mathcal{M})$ , define a stochastic choice rule  $\rho_\mu : \Omega \rightarrow [0, 1]$  such that for all trials  $(x, A) \in \Omega$ ,

$$\rho_\mu(x, A) = \mu_A(\mathcal{T}(x, A)) \quad (10)$$

is the likelihood that  $\mu_A$  assigns to all total types  $\theta$  for which  $x$  is best in  $A$ . To check that  $\rho_\mu$  is a stochastic choice rule, use the partition (3) for any menu  $A \in \mathcal{M}$ :

$$\sum_{x \in A} \rho(x, A) = \sum_{x \in A} \mu_A(\mathcal{T}(x, A)) = \mu_A(\mathcal{T}) = 1.$$

Say that a stochastic choice rule  $\rho$  is represented by a discrete map  $\mu \in \mathcal{D}(\mathcal{M})$  if  $\rho = \rho_\mu$ . This representation is called the *discrete-map expected utility* (DMEU). Obviously, DREU is a special case of DMEU where the discrete map  $\mu$  is constant.

As all distributions  $\mu_A$  share the same support  $\Theta_\mu$ , then DMEU still implies (9) and hence, Discreteness. On the other hand, DMEU violates Additivity because it puts distinct distributions  $\mu_A$  and  $\mu_{\gamma A + (1-\gamma)B}$  on the two sides of the partition (4). Similarly, Monotonicity can fail because the distributions  $\mu_A$  and  $\mu_{A \cup B}$  need not be the same.

Instead, DMEU implies a weaker form of Additivity. Let  $\{0 < \gamma_n \leq 1\}$  and  $\{B_n \in \mathcal{M}\}$  denote sequences  $\gamma_1, \gamma_2, \dots$  and  $B_1, B_2, \dots$  such that  $\gamma_n \in (0, 1]$  and  $B_n \in \mathcal{M}$  for all  $n \in \mathbb{N}$ .

**Axiom 4** (Slack Additivity (SA)). *For all trials  $(x, A) \in \Omega$  and convergent sequences  $\{0 < \gamma_n \leq 1\}$  and  $\{B_n \in \mathcal{M}\}$ ,*

$$\rho(x, A) > 0 \quad \Leftrightarrow \quad \lim_{n \rightarrow \infty} \rho(x, A, \gamma_n, B_n) > 0.$$

Given Additivity, SA is trivial because  $\rho(x, A, \gamma_n, B_n) = \rho(x, A)$  for all  $n$ . To interpret SA more broadly, suppose that the sequences  $\gamma_n = \gamma \in (0, 1]$  and  $B_n = B \in \mathcal{M}$  are constant for all  $n \in \mathbb{N}$ . Then SA reduces to the equivalence

$$\rho(x, A) > 0 \quad \Leftrightarrow \quad \rho(x, A, \gamma, B) > 0,$$

which can be also rewritten as

$$\rho(x, A) = 0 \quad \Leftrightarrow \quad \rho(x, A, \gamma, B) = 0.$$

Thus SA requires Additivity, but only in the boundary case when either side in the equation  $\rho(x, A) = \rho(x, A, \gamma, B)$  is zero.

In the general asymptotic scenario when the converging sequences  $\{0 < \gamma_n \leq 1\}$  and  $\{B_n \in \mathcal{M}\}$  are not constant, SA follows from DMEU because the discrete map  $\mu : \mathcal{D} \rightarrow \mathcal{M}$  is continuous.<sup>2</sup>

**Theorem 3.** *A stochastic choice rule  $\rho$  satisfies Discreteness and SA if and only if  $\rho = \rho_\mu$  for some discrete map  $\mu \in \mathcal{D}(\mathcal{M})$ . Such  $\mu$  is unique, and*

$$|\Theta_\mu| = \max_{A \in \mathcal{M}} \kappa_\rho(A).$$

This is our main characterization result for stochastic choice rules. It delivers various applications, such as DREU and the self-selection model below. The existence and uniqueness claims in Theorem 3 assert that the transformation

$$\mu \leftrightarrow \rho_\mu$$

is a bijection between discrete maps  $\mu \in \mathcal{D}(\mathcal{M})$  and stochastic choice rules  $\rho = \rho_\mu$  that satisfy Discreteness and SA. The size  $|\Theta_\mu|$  of the support  $\Theta_\mu$  can be achieved as the maximal number of elements chosen with positive probability in some menu. A fortiori, it is also achievable in DREU.

The construction of  $\mu$  in the proof of Theorem 3 proceeds as follows. Assume that  $\rho$  satisfies Discreteness and SA. Then there is a maximal menu  $C = \{c_1, \dots, c_k\}$  such that  $\rho(c_i, C) > 0$  for all  $i$ . For any  $x, y \in X$  and  $i = 1, \dots, k$ , let

$$x\theta_i y \quad \Leftrightarrow \quad \rho(\alpha x + (1 - \alpha)c_i, \alpha\{x, y\} + (1 - \alpha)C) > 0, \quad (11)$$

for sufficiently small  $\alpha > 0$ . Let  $\Theta_\mu = \{\theta_1, \dots, \theta_k\}$ . Take any menu  $A \in \mathcal{M}$ . For all  $\gamma \in (0, 1]$ , let  $A_\gamma = \gamma C + (1 - \gamma)A$ . For all  $\theta \in \Theta_\mu$ , let

$$\mu_A(\theta) = \lim_{\gamma \rightarrow 0} \rho(\text{best for } \theta \text{ in } A_\gamma, A_\gamma) \quad (12)$$

be the asymptotic likelihood that the best choice for the type  $\theta$  in  $A_\gamma$  is chosen in  $A_\gamma$ . For

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<sup>2</sup>See appendix for more details in this claim.

all  $\Psi \subset \mathcal{T}$ , let  $\mu_A(\Psi) = \sum_{\theta \in \Psi \cap \Theta_\mu} \mu_A(\theta)$ . Then  $\mu \in \mathcal{D}(\mathcal{M})$  is a discrete map, and  $\rho = \rho_\mu$ . All details are in the appendix.

### 3.2 Self-Selection

*Self-selection* is a form of context dependence with normative rather than behavioral incentives. Such pattern can arise when agents of any particular type  $\theta$  can increase their participation across distinct menus when their best choices are improved. For example, it should be reasonable for gamblers to bet more actively against bookmakers who offer them more favorable odds. Unlike the behavioral heuristics, like reason-based choice or extreme-ness aversion, self-selection should be consistent with Monotonicity. Indeed, all types that choose  $x$  in  $A \cup B$  should still choose  $x$  in  $A$ , and their participation rates should be the same in the two menus, while all other types should participate in  $A \cup B$  at least as much as in  $A$ .

A discrete map  $\mu \in \mathcal{D}(\mathcal{M})$  is called *selective* if for all menus  $A, B \in \mathcal{M}$  and types  $\theta \in \mathcal{T}$ ,

$$\text{best for } \theta \text{ in } A \cup B \in A \quad \Rightarrow \quad \mu_A(\theta) \geq \mu_{A \cup B}(\theta). \quad (13)$$

Note that the left-hand side implies that the type  $\theta$  should have the same best choices in menus  $A$  and  $A \cup B$ . Thus condition (13) is plausible under self-selection because such  $\theta$  should be equally motivated to participate in menus  $A$  and  $A \cup B$ , while any other possible type  $\psi$  should participate in  $A \cup B$  at least as often as in  $A$  because  $b(\psi, A)$  is also feasible in  $A \cup B$ .

Note that if  $\mu$  is selective, then  $\rho_\mu$  satisfies Monotonicity. Indeed, for any  $(x, A) \in \Omega$  and  $B \in \mathcal{M}$ , the inclusion  $\mathcal{T}(x, A) \supset \mathcal{T}(x, A \cup B)$  holds. Thus,

$$\begin{aligned} \rho_\mu(x, A) &= \mu_A(\mathcal{T}(x, A)) \geq \mu_A(\mathcal{T}(x, A \cup B)) = \\ &= \sum_{\theta \in \Theta_\mu: x=b(\theta, A \cup B)} \mu_A(\theta) \geq \sum_{\theta \in \Theta_\mu: x=b(\theta, A \cup B)} \mu_{A \cup B}(\theta) = \rho_\mu(x, A \cup B). \end{aligned}$$

Conversely, if  $\rho_\mu$  satisfies Monotonicity, then  $\mu$  is selective. Given Theorem 3, this finding can be written as

**Theorem 4.** *A stochastic choice rule  $\rho$  satisfies Discreteness, Monotonicity, and SA if and only if  $\rho = \rho_\mu$  for some selective discrete map  $\mu$ .*

This result asserts that all stochastic choice rules  $\rho = \rho_\mu$  that are represented by discrete maps  $\mu \in \mathcal{D}(\mathcal{M})$  and satisfy Monotonicity can be interpreted in terms of self-selection.

The selective property (13) is insufficient for the DMEU. In fact, some  $\rho = \rho_\mu$  where  $\mu$  is selective can fail to have a random utility representation via negative Block–Marschak polynomials. Take  $k$  types each with a utility function  $u_\theta : X \rightarrow \mathbb{R}_-$ . Let

$$\mu_A(\theta) = \frac{\sum_{\psi \neq \theta} \max_{x \in A} u_\psi(x)}{(k-1) \sum_{\psi} \max_{x \in A} u_\psi(x)}.$$

This discrete map is selective as expanding the menu without adding a better option for type  $\theta$  will decrease the numerator in magnitude faster than the denominator. Consider the following types and choice data:

| type $i$ | $u_i(a)$ | $u_i(b)$ | $u_i(c)$ | $u_i(d)$ | $\mu_{\{a,b\}}(i)$ | $\mu_{\{a,b,c\}}(i)$ | $\mu_{\{a,b,d\}}(i)$ | $\mu_{\{a,b,c,d\}}(i)$ |
|----------|----------|----------|----------|----------|--------------------|----------------------|----------------------|------------------------|
| 1        | −2       | −3       | −4       | −5       | $\frac{1}{3}$      | 0.3                  | 0.3                  | $\frac{2}{8}$          |
| 2        | −3       | −2       | −1       | −4       | $\frac{1}{3}$      | 0.4                  | 0.3                  | $\frac{3}{8}$          |
| 3        | −3       | −2       | −4       | −1       | $\frac{1}{3}$      | 0.3                  | 0.4                  | $\frac{3}{8}$          |

We now compute the Block–Marschak polynomial  $q(a, \{ab\})$

$$= \rho(a, \{a, b\}) - \rho(a, \{a, b, c\}) - \rho(a, \{a, b, d\}) + \rho(a, \{a, b, c, d\}) = \frac{1}{3} - 0.3 - 0.3 + \frac{2}{8} = \frac{-1}{60} < 0.$$

Our model of self-selection can be applied as an extension of the Luce model of stochastic choice. To do so, let  $X = \Delta^Q$  consist of lotteries on a finite set  $Q$  of deterministic outcomes. Say that  $\mu$  is a Luce discrete map if for all  $A \in \mathcal{M}$  and  $\theta \in \Theta_\mu$ ,

$$\mu_A(\theta) = \frac{\max_{x \in A} u_\theta(x)}{\sum_{\psi \in \Theta_\mu} \max_{x \in A} u_\psi(x)},$$

and  $u_\theta : X \rightarrow \mathbb{R}_{++}$  is some function that is maximized by  $\theta$ .

Let  $\rho|Q$  be the restriction of  $\rho$  to menus  $A \subset Q$ .

**Theorem 5.** *If  $\rho = \rho_\mu$  where  $\mu$  is a Luce discrete map, then  $\rho|Q$  satisfies the random utility model.*

When a stochastic choice rule is generated by a Luce discrete map, the random utility model need not uniquely identify the menu-invariant distribution of types. For example, let  $|X| = 5$ . Each trial (i.e., menu-element pair) generates a linear equality relating observed choice probabilities to the distribution over the  $5! = 120$  strict preference orderings. The total number of such equalities is

$$\sum_{k=0}^5 \binom{5}{5-k} \times (5-k) = 5 \times 2^{5-1} = 80.$$

Since the number of variables (120) exceeds the number of equalities (80),<sup>3</sup> the distribution over preference types is not uniquely determined. This conclusion extends to any  $|X| \geq 5$  as the  $|X|!$  grows faster than  $|X| \times 2^{|X|-1}$ . This conclusion is consistent with Turansick (2022), who shows that for  $|X| > 4$ , any random utility representation of a choice system satisfying Luce<sup>4</sup> is not unique. By Theorem 3, our DMEU model avoids this issue and returns a unique distribution, given some support  $\Theta_\mu$ , for a  $\rho$  that satisfies Discreteness and SA.

Our Luce discrete map joins many extensions to the standard Luce model (Luce, 1959) in recent literature. Gul, Natenzon, and Pesendorfer (2014) relates  $\rho(x, A)$  to the proportion of the values of attributes  $x$  has relative to those of all attributes observed in a menu. Agents hence view each option as a bundle of attributes. Echenique and Saito (2019) captures just-noticeable differences by allowing dominated items still be chosen with positive probability when compared to closer alternatives. Kovach and Tserenjigmid (2022) touches upon limited attention by weighing items higher if they are in the focal set of the menu. This model also discusses menu dependence like in our DMEU.

## 4 Discussion

DREU can be naturally compared with GP's random expected utility. In their model,  $X = \Delta^n$  is a finite-dimensional simplex, and for all  $(x, A) \in \Omega$ ,

$$\rho(x, A) = \mu(\mathcal{C}(x, A)), \quad (14)$$

where  $\mu$  is a finitely additive probability measure over  $\mathcal{C}$ , and

$$\mathcal{C}(x, A) = \{\theta \in \mathcal{C} : x \text{ is best for } \theta \text{ in } A\}.$$

Note that  $\mathcal{C} = \bigcup_{x \in A} \mathcal{C}(x, A)$ , but this union need not be disjoint because a mixture continuous type  $\theta \in \mathcal{C}$  can have several best elements in  $A$ . To make representation (14) consistent with normalization (7), it must be assumed that the probability of ties is zero in each menu  $A$ , that is,

$$\mu(\mathcal{C}(x, A) \cap \mathcal{C}(y, A)) = 0 \quad (15)$$

for all distinct  $x, y \in A$ . GP calls such distributions  $\mu$  *regular*. Regularity prohibits  $\mu$  to have a finite support, except for the case when  $X$  is collinear and there are only two possible types.

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<sup>3</sup>In fact, some equalities are linearly dependent.

<sup>4</sup>The standard Luce model is a special case of our Luce discrete map.

By contrast, our representation (8) uses total rather than mixture continuous types, and takes  $\mu$  to be discrete. This change in mathematical structure allows to identify  $\mu$  more directly via the algebraic formula (11) rather than GP's analytical arguments. Our approach is more flexible about the consumption domain:  $X$  is convex, but need not be closed, or bounded, or finite dimensional. For example,  $X$  may consist of monetary gambles with payoffs on the real line, or  $X$  can be the non-negative orthant of consumption bundles.

GP's REU implies Additivity. Similarly to partition (4) for total types, one can derive from (14) that for all  $(x, A) \in \Omega$ ,  $B \in \mathcal{M}$ , and  $\gamma \in (0, 1]$ .

$$\mathcal{C}(x, A) = \bigcup_{z \in [\gamma x + (1-\gamma)B]} \mathcal{C}(z, \gamma A + (1-\gamma)B),$$

but this union need not be disjoint because best elements are not unique for mixture continuous types. Additivity still follows from representation (14) that applies  $\mu$  to both sides of the above union and assumes the no-tie constraint (15).

On the other hand, our DREU implies all axioms in GP's Theorem 2. For any menu  $A \in \mathcal{M}$ , let  $\rho_A$  be the discrete distribution over  $X$  such that  $\rho_A(x) = \rho(x, A)$  if  $x \in A$  and  $\rho_A(x) = 0$  if  $x \notin A$ . Endow the space of such distributions with the topology of weak convergence. Besides Monotonicity, GP's Theorem 2 requires the following conditions for all  $A, B \in \mathcal{M}$ ,  $x, y \in X$ , and  $\gamma \in [0, 1]$ ,

- (i) Linearity: if  $\gamma > 0$  and  $x \in A$ , then  $\rho(x, A) = \rho(x, A, \gamma, y)$ ,
- (ii) Mixture Continuity:  $\rho_{\gamma A + (1-\gamma)B}$  is continuous with respect to variations in  $\gamma$ ,
- (iii) Extremeness: if  $x \neq y$  and  $\gamma \in (0, 1)$ , then

$$\rho(\gamma x + (1-\gamma)y, \{x, y, \gamma x + (1-\gamma)y\}) = 0.$$

Linearity is a special case of Additivity where  $B = y$  is a singleton. Both MC and Extremeness are implied by DMEU and a fortiori, by DREU. Indeed, the mixture  $\gamma x + (1-\gamma)y$  where  $\gamma \in (0, 1)$  cannot be best for any total type  $\theta$  in a menu  $A$  that contains both  $x$  and  $y$ . Given Monotonicity, Extremeness implies that if  $z$  is chosen with a positive probability in a menu  $A$ , then  $z$  is an extreme point in  $A$ . In the proof of Theorem 3, we derive MC from (10).

Therefore, if  $X = \Delta^n$ , and Discreteness holds, then the lists of axioms in our Theorem 2 and GP's Theorem 2 become equivalent. By contrast, their Theorem 3 requires that  $\mu$  in (14) is countably additive, and  $\rho_A$  is continuous with respect to  $A$ . Such continuity is

incompatible with DMEU because the stochastic choice rules  $\rho_\theta$  are not continuous for total types  $\theta$ , except when  $X = \Delta^2$ .

We conjecture that Additivity can be derived from GP's axioms even if  $X$  is not finite dimensional, but proving this claim appears complicated and tangential to our desiderata. So we skip this effort. Besides making the proofs and the statement of Theorem 2 more parsimonious, Additivity is conveniently relaxed to SA in Theorem 3, which is conceptually our main result.

Note also that Additivity can be derived from its subadditivity part: for all  $(x, A) \in \Omega$ ,  $B \in \mathcal{M}$ , and  $\gamma \in (0, 1)$ ,

$$\rho(x, A) \geq \rho(x, A, \gamma, B).$$

Indeed, if any of these inequalities is strict, then by (7),

$$1 = \sum_{x \in A} \rho(x, A) > \sum_{x \in A} \sum_{y \in B} \rho(\gamma x + (1 - \gamma)y, \gamma A + (1 - \gamma)B) \geq 1.$$

By contradiction, Additivity holds.

## 4.1 Random Risk Aversion

Olschewski et al. (2022) discovered that the random utility model does not precisely differentiate between risk-averse and risk-seeking agents. This motivates us to next extend the discussion of menu dependence to risk attitudes.

Let  $X = \Delta^Q$  be the set of all lotteries on some interval  $Q \subset \mathbb{R}$  of monetary prizes.<sup>5</sup> This interval is convex, but need not be closed or bounded. Any  $q \in Q$  is associated with the degenerate lottery that delivers  $q$  with certainty. Accordingly,  $Q \subset X$  denotes the subclass of such degenerate lotteries.

By mixture linearity, any utility function  $u \in \mathcal{U}$  has the expected utility form:

$$u(x) = \sum_{q \in Q} x(q)u(q) \quad \text{for all } x \in X.$$

The restriction of  $u$  to  $Q$  is known as the *Bernoulli utility index*, and its concavity can be interpreted in terms of *risk aversion*. Formally,  $u$  is strictly increasing and concave on  $Q$  if

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<sup>5</sup>This example can be extended to the set  $X$  of all Borel distributions on  $Q$  with bounded rather than finite support. For brevity, we adopt the simplest framework here that still allows to identify a random risk aversion model.

and only if for all  $x \in X$  and  $q \in Q$ ,

$$q > E(x) \quad \Rightarrow \quad q > u(x),$$

where  $E(x) = \sum_{q \in Q} x(q)q$  is the mathematical expectation (i.e. the mean) of the lottery  $x$ . Risk aversion can be imposed on total types in our model as well.

A type  $\theta \in \mathcal{T}$  is called *risk averse* if it maximizes some  $u \in \mathcal{U}$  with a strictly increasing and concave restriction to  $Q$ . Let  $\mathcal{T}_{ra}$  be the set of all risk averse types. Note that the concavity is not required to be strict and hence, the risk neutral types  $\theta$  that maximize  $u(x) = E(x)$  are included in  $\mathcal{T}_{ra}$ .

**Axiom 5** (Archimedean Risk Aversion (ARA)). *For any  $q \in Q$  and  $x, y \in \Delta^Q$ , there is  $\varepsilon > 0$  such that*

$$q > E(x) \quad \Rightarrow \quad \rho(x, \{x, \varepsilon y + (1 - \varepsilon)q\}) = 0.$$

When  $y = q$ , then ARA implies

$$q > E(x) \quad \Rightarrow \quad \rho(x, \{x, q\}) = 0.$$

This part assumes that any type  $\theta \in \Theta_\mu$  should prefer the certain payoff  $q$  to any gamble  $x$  that delivers less than  $q$  on average. The Archimedean extension is necessary to guarantee that the strict risk aversion of the type  $\theta$  should be reflected by the expected utility function  $u$  that is maximized by  $\theta$  rather than the tie-breaking rule that is embedded in  $\theta$ .

**Corollary 6.** *For any discrete map  $\mu \in \mathcal{D}(\mathcal{M})$ ,  $\rho_\mu$  satisfies ARA if and only if  $\Theta_\mu \subset \mathcal{T}_{ra}$ .*

This result is derived from Theorem 1. Corollary 6 can be combined with Theorems 2–4 to characterize DMEU and its refinements where all types are required to be risk averse.

Peng (2026) explores the effects of a certain option (degenerate lottery) on risk behavior in an experimental setting. Pilot results show violations of Monotonicity. Hence, GP’s standard REU would fail to model this phenomenon, and menu-dependence is needed. This phenomenon can be explained by treating the certain option as a decoy, where its presence will make some agents more risk averse.

This can be extended to more types. Specifically, we can use discrete Arrow-Pratt indices by Baillon and L’Haridon (2021) to capture the finite nature of the type space. A discrete lognormal distribution may be appropriate for  $\mu_A(\cdot)$  as CRRA, in the format of  $u(x) = \frac{x^r}{1-r}$ , indices are positive and skewed right, with most agents exhibiting some level of risk aversion. The mode of  $\mu_A$  will move left (towards 0) when a certain option is present. Desmettre and Steffensen (2023) also used random risk aversion in CARA and CRRA to solve an investor’s portfolio allocation problem.

## 4.2 Random Beliefs and Cobb-Douglas Utility

To accommodate randomness in beliefs rather than risk attitudes, let  $X = Q^n$  for some interval  $Q \subset \mathbb{R}$ . Interpret each  $x \in X$  as an uncertain prospect that pays a monetary reward  $x(s)$  contingent on a realized state of nature  $s \in \{1, \dots, n\}$ .

A mixture linear utility function  $u \in \mathcal{U}$  is called *positive* if for all  $x \in X$ ,

$$u(x) = \sum_{s=1}^n p_s x(s)$$

for some  $p_1, \dots, p_n \in \mathbb{R}_+$  such that  $\sum_{s=1}^n p_s = 1$ . As  $u$  is not constant, then  $p_s > 0$  for some  $s \in \{1, \dots, n\}$ . The weights  $p_s$  are interpreted as subjective beliefs attached to the states  $s \in \{1, \dots, n\}$ .

Alternatively, one can take  $Q = \mathbb{R}_{++}$  and interpret each vector  $x$  in the positive orthant  $X = \mathbb{R}_{++}^n$  as a consumption bundle in the logarithmic form so that for each index  $s \in \{1, \dots, n\}$ , good  $s$  is available in quantity  $e^{x(s)}$  in the bundle  $x$ . Then each positive utility function has the *Cobb-Douglas* form, which is one of the most common utility models used across economic applications. Apesteguia and Ballester (2023) discusses identification and estimation methods for random Cobb-Douglas, along with other empirically relevant frameworks. Cobb-Douglas production functions, when taken logs, can be treated as a linear regression model with random coefficients  $y = a + \mathbf{x}'b$  (Fox and Gandhi, 2016).

A type  $\theta \in \mathcal{T}$  is called *positive* if it maximizes some positive mixture linear function  $u \in \mathcal{U}$ . Let  $\mathcal{T}_p$  be the set of all positive types.

**Axiom 6** (Positivity). *For any bundles  $x, y \in \mathbb{R}_{++}^n$ , if  $x_i > y_i$  for all  $i = 1, \dots, n$ , then  $\rho(x, \{x, y\}) = 1$ .*

This axiom requires that all types  $\theta$  that choose between  $x$  and  $y$  should satisfy a standard notion of non-satiation for consumption bundles.

**Corollary 7.** *For any discrete map  $\mu \in \mathcal{D}(\mathcal{M})$ ,  $\rho_\mu$  satisfies Positivity if and only if  $\Theta_\mu \subset \mathcal{T}_p$ .*

This result is a consequence of Theorem 1, and can be combined with Theorems 2–4. In this way, one can study the randomness of Cobb-Douglas utility functions in terms of stochastic choice data in finite (e.g. binary) menus. The key axioms of Additivity and SA for this application are illustrated as follows in Figure 1.

Let each color represent a menu. The green menu mixes the red with blue at a proportion of  $\frac{2}{3} : \frac{1}{3}$ . We can observe Additivity of an item from a red menu, say  $+$ , if the distribution that chooses it gets partitioned into  $\oplus$  and  $\boxplus$ .

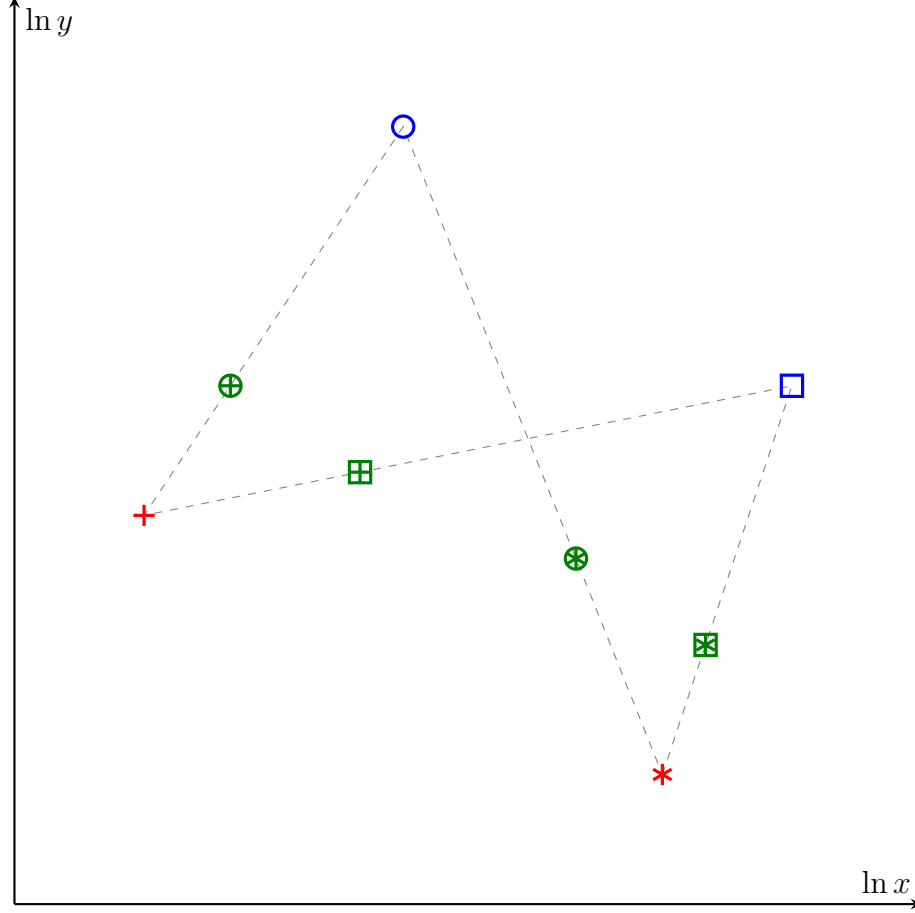


Figure 1: The green menu is a mixture of the red and blue menus at  $\frac{2}{3} : \frac{1}{3}$ .

Randomness in beliefs in DMEU can be used to study learning patterns where beliefs can depend on the composition of the menu  $A$ . Another possible extension is to accommodate randomness in beliefs and risk attitudes together. To do so, one can take  $X$  to be a set  $\Delta^{Q \times S}$  of all Anscombe and Aumann's (1963) acts—functions  $x : S \rightarrow \Delta^Q$  on a finite state space  $S$ . This domain can be embedded as a convex compact subset of  $L = \mathbb{R}^{Q \times S}$ . We delegate the discussion of such extensions to a companion paper.

### 4.3 Tie-Breakers

The mixture linear utility maximization in Theorem 1 can be refined with a more explicit description of tie-breaking rules. For any  $u \in \mathcal{U}$ , let

$$L_u = \{\alpha(x - y) : \alpha \geq 0 \text{ and } x, y \in X \text{ such that } u(x) = u(y)\}.$$

As  $X$  is convex, then  $L_u$  is a linear space.<sup>6</sup>

**Theorem 8.** *A type  $\theta$  is stable if and only if there is a function  $u \in \mathcal{U}$  and a type  $\psi$  in  $L_u$  such that for all  $x, y \in X$ ,*

$$x\theta y \Leftrightarrow \begin{aligned} &u(x) > u(y), \text{ or} \\ &u(x) = u(y) \text{ and } (x - y)\psi 0. \end{aligned} \quad (16)$$

If (16) holds with another pair  $(u', \psi')$ , then  $u' = \alpha u + \beta$  for some  $\alpha > 0$  and  $\beta \in \mathbb{R}$ ,  $L_u = L_{u'}$ , and  $\psi = \psi'$ .

Representation (16) asserts that any stable type  $\theta$  maximizes some mixture linear utility function  $u \in \mathcal{U}$ , which is unique up to a positive linear transformation, and then uses the unique residual type  $\psi$  on the linear subspace  $L_u$  to compare any  $x, y \in X$  such that  $u(x) = u(y)$ . Moreover,  $\theta$  is total if and only if  $\psi$  is total.

For any  $u \in \mathcal{U}$ , let  $\mathcal{T}_u$  be the set of all total types on  $L_u$ . Such types are called *tie-breakers*. To construct tie-breakers, take any *basis*  $H$  in the linear space  $L_u$ . By definition,  $H \subset L_u$  is a linearly independent system of vectors that spans  $L_u$ . This definition is standard if  $L_u$  is finite dimensional. More generally, such  $H$  is known as a *Hamel basis* when  $L_u$  is infinite dimensional. The existence of a Hamel basis in any linear space is guaranteed by Zorn's lemma (see Aliprantis and Border (1999, Theorem 1.6)). For any  $x \in L_u$  and  $h \in H$ , let  $x(h)$  be the coefficient (coordinate) of the vector  $h$  in the unique linear combination of basis vectors that equals  $x$ . Endow  $H$  with any total order  $\geq$ . Define a binary relation  $\psi_H$  for all  $x, y \in L_u$  via

$$x\psi_H y \Leftrightarrow x = y \text{ or } x(h(x, y)) \geq y(h(x, y)), \quad (17)$$

where  $h(x, y) \in H$  is the first vector in  $H$  such that  $x(h) \neq y(h)$ . Then  $\psi_H$  compares the coordinates of the vectors  $x$  and  $y$  *lexicographically* in the basis  $H$ . By definition,  $\psi_H$  is a tie-breaker: it is complete, total, transitive, and satisfies Independence.

Thus a rich family of total types can be generated by representations (16) and (17). These representations can be further refined (e.g. Fishburn (1982), Myerson (1986), Blume, Brandenburger, and Dekel (1991)) by imposing more structure on the consumption domain  $X$ , the functions  $u \in \mathcal{U}$ , and the basis  $H$ .

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<sup>6</sup>Indeed, take any  $\alpha \geq 0$  and  $x, y \in X$  such that  $u(x) = u(y)$ . Then  $\gamma\alpha(x - y) \in L_u$  for all  $\gamma \in \mathbb{R}$  because if  $\gamma < 0$ , then  $\gamma\alpha(x - y) = (-\gamma\alpha)(y - x)$ . For any  $\beta \geq 0$  and  $x', y' \in X$  such that  $u(x') = u(y')$ , the sum

$$\alpha(x - y) + \beta(x' - y') = (\alpha + \beta) \left[ \left( \frac{\alpha}{\alpha + \beta}x + \frac{\beta}{\alpha + \beta}x' \right) - \left( \frac{\alpha}{\alpha + \beta}y + \frac{\beta}{\alpha + \beta}y' \right) \right]$$

also belongs to  $L_u$  because  $u \left( \frac{\alpha}{\alpha + \beta}x + \frac{\beta}{\alpha + \beta}x' \right) = u \left( \frac{\alpha}{\alpha + \beta}y + \frac{\beta}{\alpha + \beta}y' \right)$ .

## A APPENDIX: PROOFS

First, we show that a discrete map  $\mu \in \mathcal{D}(\mathcal{M})$  can be derived uniquely from the stochastic choice rule  $\rho_\mu$ . Then this identification procedure is used to prove Theorems 2, 3, and 4.

To derive  $\mu$  from  $\rho_\mu$ , we need some preliminary steps.

**Lemma A.1.** *For any menu  $A \in \mathcal{M}$  and converging sequences  $\{0 < \gamma_n \leq 1\}$  and  $\{B_n \in \mathcal{M}\}$  such that  $\gamma = \lim_{n \rightarrow \infty} \gamma_n$  and  $B = \lim_{n \rightarrow \infty} B_n$ ,*

$$\lim_{n \rightarrow \infty} \gamma_n A + (1 - \gamma_n) B_n = \gamma A + (1 - \gamma) B.$$

*Proof.* Take any  $A \in \mathcal{M}$  and converging sequences  $\{0 < \gamma_n \leq 1\}$  and  $\{B_n \in \mathcal{M}\}$  such that  $\gamma = \lim_{n \rightarrow \infty} \gamma_n$  and  $B = \lim_{n \rightarrow \infty} B_n$ . Show that for all  $n \in \mathbb{N}$ ,

$$d_h(\gamma A + (1 - \gamma) B, \gamma_n A + (1 - \gamma_n) B_n) \leq R|\gamma - \gamma_n| + d_h(B, B_n) \quad (18)$$

where  $R = \max_{x \in A, y \in B} \|x - y\|$ . Take any  $z \in \gamma A + (1 - \gamma) B$ . Then  $z = \gamma x + (1 - \gamma)y$  for some  $x \in A$  and  $y \in B$ . By the definition of the Hausdorff metric  $d_h$ , there is  $y_n \in B_n$  such that  $\|y - y_n\| \leq d_h(B, B_n)$ . Let  $z_n = \gamma_n x + (1 - \gamma_n)y_n$ . Then  $z_n \in \gamma_n A + (1 - \gamma_n) B_n$ , and

$$\begin{aligned} \|z - z_n\| &\leq \|z - (\gamma_n x + (1 - \gamma_n)y)\| + \|(\gamma_n x + (1 - \gamma_n)y) - z_n\| \leq \\ &|\gamma - \gamma_n| \|x - y\| + \gamma_n \|y - y_n\| \leq R|\gamma - \gamma_n| + d_h(B, B_n). \end{aligned}$$

Similarly, for any  $z_n \in \gamma_n A + (1 - \gamma_n) B_n$ , there is  $z \in \gamma A + (1 - \gamma) B$  such that

$$\|z - z_n\| \leq R|\gamma - \gamma_n| + d_h(B, B_n).$$

Thus (18) holds. By (18),  $\lim_{n \rightarrow \infty} \gamma_n A + (1 - \gamma_n) B_n = \gamma A + (1 - \gamma) B$ . □

For any menu  $C \in \mathcal{M}$ , define a function  $b_C : \mathcal{T} \rightarrow C$  such that

$$b_C(\theta) = b(\theta, C) \quad \text{for all } \theta \in \mathcal{T}.$$

Say that  $b_C$  is bijective on  $\Theta \subset \mathcal{T}$  if for any  $x \in C$ , there is a unique  $\theta \in \Theta$  such that  $x = b_C(\theta)$ .

**Lemma A.2.** *For any finite set  $\Theta \subset \mathcal{T}$ , there is  $C \in \mathcal{M}$  such  $b_C$  is bijective on  $\Theta$ .*

*Proof.* Let  $\Theta = \{\theta_1, \dots, \theta_k\}$  consist of  $k = |\Theta|$  distinct total types. Let  $I$  be the set of all pairs  $i, j \in \{1, \dots, k\}$  such that  $i > j$ . For each pair  $(i, j) \in I$ , the types  $\theta_i, \theta_j \in \mathcal{D}$

are distinct and total. Thus there are distinct elements  $x_{ij}, x_{ji} \in X$  such that  $x_{ij}\theta_i^*x_j$  and  $x_{ji}\theta_j^*x_i$ . For any  $n \in \{1, \dots, k\}$ , let

$$c_n = \sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^n,$$

where  $x_{ij}^n = x_{ij}$  if  $x_{ij}\theta_n^*x_{ji}$  and  $x_{ij}^n = x_{ji}$  if  $x_{ji}\theta_n^*x_{ij}$ . For any  $(m, n) \in I$ , the types  $\theta_n, \theta_m$  satisfy Independence and hence,

$$\begin{aligned} c_m &= \left( \sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^m \right) \theta_m^* \left( \sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^n \right) = c_n \\ c_n &= \left( \sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^n \right) \theta_n^* \left( \sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^m \right) = c_m \end{aligned} \tag{19}$$

because  $x_{ij}^n \theta_n x_{ij}^m$  and  $x_{ij}^m \theta_m x_{ij}^n$  for all  $(i, j) \in I$ , and  $x_{ij}^n \theta_n^* x_{ij}^m$  and  $x_{ij}^m \theta_m^* x_{ij}^n$  when  $(i, j) = (m, n)$ . Let  $C = \{c_1, \dots, c_k\}$ . By (19), for all  $i = 1, \dots, k$ ,  $c_i = b_C(\theta_i)$ . Thus  $b_C$  is bijective on  $\Theta$ .  $\square$

Take any discrete map  $\mu \in \mathcal{D}(\mathcal{M})$  and any finite set  $\Theta \subset \mathcal{T}$  such that  $\Theta_\mu \subset \Theta$ . By Lemma A.2, there is a menu  $C \in \mathcal{M}$  such that  $b_C$  is bijective on  $\Theta$ . For any  $\gamma \in (0, 1]$  and  $A \in \mathcal{M}$ , let

$$A_\gamma = \gamma C + (1 - \gamma)A.$$

Show that for all  $\theta \in \Theta$ ,

$$\mu_{A_\gamma}(\theta) = \rho_\mu(b(\theta, A_\gamma), A_\gamma). \tag{20}$$

Take any  $\psi \in \Theta_\mu$  such that  $\psi \neq \theta$ . As  $b_C$  is bijective on  $\Theta_\mu$ , then  $b(\theta, C) \neq b(\psi, C)$  and hence,  $b(\theta, C)\theta^*b(\psi, C)$  because  $\theta$  is total. By Independence,

$$b(\theta, A_\gamma) = [\gamma b(\theta, C) + (1 - \gamma)b(\theta, A)]\theta^*[\gamma b(\psi, C) + (1 - \gamma)b(\psi, A)] = b(\psi, A_\gamma).$$

As  $\theta$  is the only type in  $\Theta$  for which  $b(\theta, A_\gamma)$  is best in  $A_\gamma$ , then (20) holds by definition of  $\rho_\mu$ .

As  $\mu$  is continuous, then by Lemma A.1, for all  $A \in \mathcal{M}$  and  $\theta \in \Theta$ ,

$$\mu_A(\theta) = \lim_{\gamma \rightarrow 0} \mu_{A_\gamma}(\theta) = \lim_{\gamma \rightarrow 0} \rho_\mu(b(\theta, A_\gamma), A_\gamma). \tag{21}$$

This identification formula has several implications in our results.

1. First,  $\mu$  is determined uniquely by  $\rho_\mu$ . Indeed, if  $\mu' \in \mathcal{D}(\mathcal{M})$  is such that  $\rho_\mu = \rho_{\mu'}$ ,

then by (21), for all menus  $A \in \mathcal{M}$  and types  $\theta \in \Theta = \Theta_\mu \cup \Theta_{\mu'}$ ,

$$\mu_A(\theta) = \lim_{\gamma \rightarrow 0} \rho_\mu(b(\theta, A_\gamma), A_\gamma) = \mu'_A(\theta).$$

Thus  $\mu = \mu'$ .

2. Second, if  $\rho_\mu$  satisfies Additivity, then  $\mu \in \mathcal{D}$  is constant. Let  $\Theta = \Theta_\mu$ . By Additivity, for all  $\theta \in \Theta_\mu$  and  $\gamma \in (0, 1]$

$$\rho_\mu(b(\theta, C), C) = \rho_\mu(b(\theta, C), C, \gamma, A) \geq \rho_\mu(b(\theta, A_\gamma), A_\gamma)$$

because  $b(\theta, A_\gamma) \in \gamma b(\theta, C) + (1 - \gamma)A$ . As

$$\sum_{\theta \in \Theta} \rho_\mu(b(\theta, C), C) = \sum_{\theta \in \Theta} \rho_\mu(b(\theta, A_\gamma), A_\gamma) = 1$$

then  $\rho_\mu(b(\theta, C), C) = \rho_\mu(b(\theta, A_\gamma), A_\gamma)$  for all  $\theta \in \Theta$ . By (21),  $\mu_A(\theta) = \rho_\mu(b(\theta, C), C)$  is invariant of  $A$ .

3. Third, if  $\rho_\mu$  satisfies Monotonicity, then  $\mu$  is selective. Let  $\Theta = \Theta_\mu$ . Take any menus  $A, B \in \mathcal{M}$  and type  $\theta \in \Theta$  such that  $b(\theta, A \cup B) \in A$ . Then  $b(\theta, A) = b(\theta, A \cup B)$ . By Independence, for all  $\gamma \in (0, 1]$ ,

$$b(\theta, A_\gamma) = \gamma b(\theta, C) + (1 - \gamma)b(\theta, A) = \gamma b(\theta, C) + (1 - \gamma)b(\theta, A \cup B) = b(\theta, A_\gamma \cup B_\gamma).$$

By Monotonicity,

$$\rho_\mu(b(\theta, A_\gamma), A_\gamma) \geq \rho_\mu(b(\theta, A_\gamma \cup B_\gamma), A_\gamma \cup B_\gamma).$$

By (21)  $\mu_A(\theta) \geq \mu_{A \cup B}(\theta)$ . Thus  $\mu$  is selective.

We have shown already that if  $\mu$  is selective, then  $\rho_\mu$  satisfies Monotonicity, and if  $\mu$  is constant, then  $\rho_\mu$  satisfies Additivity. Thus Theorems 2 and 4 follow from Theorem 3, which we prove next.

## A.1 Proof of Theorem 3

Theorem 3 is the main step in the proofs of our representation results for stochastic choice functions  $\rho$ . So we prove it first.

Let  $\rho = \rho_\mu$  for some discrete map  $\mu \in \mathcal{D}(\mathcal{M})$  with support  $\Theta_\mu$ . The uniqueness of  $\mu$  has been established.

By (9),  $\kappa_\rho(A) \leq |\Theta_\mu|$ , and the Discreteness holds. Show SA. Take any  $(x, A) \in \Omega$  and converging sequences  $\{0 < \gamma_n \leq 1\}$  and  $\{B_n \in \mathcal{M}\}$  such that  $\gamma = \lim_{n \rightarrow \infty} \gamma_n$  and  $B = \lim_{n \rightarrow \infty} B_n$ . Then by Lemma (A.1),

$$\begin{aligned} \rho(x, A) > 0 &\Leftrightarrow \mu_A(\mathcal{T}(x, A)) > 0 \Leftrightarrow \mu_{\gamma A + (1-\gamma)B}(\mathcal{T}(x, A)) > 0 \Leftrightarrow \\ &\lim_{n \rightarrow \infty} \mu_{\gamma_n A + (1-\gamma_n)B_n}(\mathcal{T}(x, A)) > 0 \Leftrightarrow \lim_{n \rightarrow \infty} \rho(x, A, \gamma_n, B_n) > 0, \end{aligned}$$

where  $\rho(x, A, \gamma_n, B_n) = \mu_{\gamma_n A + (1-\gamma_n)B_n}(\mathcal{T}(x, A))$  by the partition (4). SA follows.

Suppose that  $\rho$  satisfies SA and Discreteness. For all  $A \in \mathcal{M}$ , let

$$\Phi(A) = \{x \in A : \rho(x, A) > 0\}$$

be the set of all elements in  $A$  that are chosen with a positive probability.

For all  $(x, A) \in \Omega$ ,  $B \in \mathcal{M}$ ,  $y \in X$ , and  $\gamma \in (0, 1]$ , SA implies

(SA1)  $\rho(x, A) > 0$  if and only if  $\rho(x, A, \gamma, B) > 0$ ,

(SA2)  $\Phi(\gamma A + (1 - \gamma)y) = \gamma\Phi(A) + (1 - \gamma)y$ ,

(SA3) if  $x \in \Phi(A \cup B)$ , then  $x \in \Phi(A)$ ,

SA1 is the restriction of SA to constant sequences  $\gamma_n = g \in (0, 1]$  and  $B_n = B \in \mathcal{M}$ . SA2 follows from SA1 by taking a singleton  $B = y$  for some  $y \in X$ ,

$$\rho(x, A) > 0 \Leftrightarrow \rho(\gamma x + (1 - \gamma)y, \gamma A + (1 - \gamma)y) > 0.$$

SA3 is the standard Chernoff property of the filter  $\Phi$ . It follows from SA1,

$$\begin{aligned} \rho(x, A \cup B) > 0 &\Rightarrow \rho(x, A \cup B, \tfrac{1}{2}, A) > 0 \Rightarrow \\ \rho(x, A, \tfrac{1}{2}, A \cup B) > 0 &\Rightarrow \rho(x, A) > 0, \end{aligned}$$

where  $\rho(x, A \cup B, \frac{1}{2}, A) \leq \rho(x, A, \frac{1}{2}, A \cup B)$  because  $\frac{1}{2}x + \frac{1}{2}A \subset \frac{1}{2}x + \frac{1}{2}(A \cup B)$ .

By Discreteness, there is a menu  $B \in \mathcal{M}$  where  $\rho$  achieves its maximal capacity across all menus,

$$K = \kappa_\rho(B) = \max_{A \in \mathcal{M}} \kappa_\rho(A).$$

Let  $C = \Phi(B)$ . By Chernoff,  $\Phi(C) = C$  and hence,  $\kappa_\rho(C) = K$ .

Write  $C$  as  $C = \{c_1, c_2, \dots, c_K\}$ .

**Lemma A.3.** *There is a finite set  $\Theta \subset \mathcal{T}$  such that  $|\Theta| = K$ , and for all  $A \in \mathcal{M}$ ,*

$$\Phi(A) = \bigcup_{\theta \in \Theta} b(\theta, A).$$

*Such  $\Theta$  is unique.*

This lemma refines the *multiple utility* model proposed by Aizerman and Malishevski Aizerman and Malishevski (1981) by restricting the type space  $\Theta$  to  $\mathcal{T}$  rather than to arbitrary total orders.

*Proof.* To show this claim, assume first that  $X$  is bounded, and there is  $R > 0$  such that  $\|x\| < R$  for all  $x \in X$ . Take  $\alpha \in (0, 1)$  such that

$$2\alpha R < (1 - \alpha) \min_{1 \leq i < j \leq K} \|c_i - c_j\|.$$

Then for any menu  $A \in \mathcal{M}$ ,

$$|\alpha A + (1 - \alpha)C| = |A|K.$$

Indeed, take any  $a, b \in A$  and  $c_i, c_j \in C$  such that  $\alpha a + (1 - \alpha)c_i = \alpha b + (1 - \alpha)c_j$ . Then

$$\alpha(a - b) = (1 - \alpha)(c_i - c_j) = 0$$

because otherwise,  $\alpha\|a - b\| \leq 2\alpha R < (1 - \alpha)\|c_i - c_j\|$  is a contradiction. Therefore all  $|A|K$  mixtures  $\alpha a + (1 - \alpha)c_i$  where  $a \in A$  and  $c_i \in C$  are distinct.

For any  $x, y \in X$ , define a binary relation  $\theta_i$  such that  $x\theta_i y$  if and only if

$$\alpha x + (1 - \alpha)c_i \in \Phi(\alpha\{x, y\} + (1 - \alpha)C).$$

By

□

## A.2 Proof of Theorem 1

Take any type  $\theta$ . Let  $\theta$  maximize  $u \in \mathcal{U}$ . Then for all  $x, y \in X$ ,

$$u(x) > u(y) \quad \Rightarrow \quad x \succ y. \tag{22}$$

As  $u$  is non-constant, then  $u(x) > u(y)$  for some  $x, y \in X$ . For any  $z_x, z_y \in X$ , take  $\gamma \in (0, 1]$  such that

$$\gamma u(z_x) + (1 - \gamma)u(x) > \gamma u(z_y) + (1 - \gamma)u(y).$$

By (22),  $\gamma z_x + (1 - \gamma)x \succ \gamma z_y + (1 - \gamma)y$ . Thus  $\theta$  is stable.

Suppose instead that  $\theta$  is stable and show that  $\theta$  maximizes some  $u \in \mathcal{U}$ . Take any menu  $A = \{a_1, a_2, \dots, a_n\}$  such that

$$a_1 \succeq a_2 \succeq \dots \succeq a_n.$$

Let  $X_A$  be the convex hull of  $A$  in  $L$ . As  $X$  is convex, then  $X_A \subset X$ . Show that for all  $z \in X_A$ ,

$$a_1 \succeq z \succeq a_n. \quad (23)$$

The proof is by induction with respect to  $n \geq 2$ . If  $n = 2$ , then by Independence, (23) holds for all  $z \in X_A$  because  $z = \gamma a_1 + (1 - \gamma)a_2$  for some  $\gamma \in [0, 1]$ . Let  $n > 2$ . Take any  $z \in X_A$  and write it as  $z = \gamma y + (1 - \gamma)a_n$  for some  $\gamma \in [0, 1]$  and  $y \in X_{A \setminus a_n}$ , where  $X_{A \setminus a_n}$  is the convex hull of  $A \setminus a_n$ . By the inductive assumption,  $a_1 \succeq y \succeq a_{n-1} \succeq a_n$ . By Independence,

$$a_1 \succeq \gamma y + (1 - \gamma)a_1 \succeq \gamma y + (1 - \gamma)a_n \succeq \gamma a_n + (1 - \gamma)a_n = a_n$$

and hence, (23) holds.

Let  $\mathcal{U}_A$  be the set of all non-constant mixture linear functions  $u : X_A \rightarrow \mathbb{R}$ .

**Lemma A.4.** *If  $a_1 \succ a_n$ , then there is  $u \in \mathcal{U}_A$  such that for all  $x, y \in X_A$ ,*

$$x \succeq y \quad \Rightarrow \quad u(x) \geq u(y). \quad (24)$$

*Moreover, such  $u$  is unique up to a positive linear transformation.*

*Proof.* Suppose that  $a_1 \succ a_n$ . For each  $\alpha \in [0, 1]$ , let  $z_\alpha = \alpha a_1 + (1 - \alpha)a_n$ . Show that for all  $\alpha, \beta \in [0, 1]$ ,

$$\alpha \geq \beta \quad \Leftrightarrow \quad z_\alpha \succeq z_\beta. \quad (25)$$

As  $a_1 \succ a_n$ , then by Independence,

$$\begin{aligned} \alpha > \beta \quad \Rightarrow \quad z_\alpha = \alpha a_1 + (1 - \alpha)a_n \succ a_n \quad \Rightarrow \\ z_\alpha = \frac{\beta}{\alpha} z_\alpha + \frac{\alpha - \beta}{\alpha} z_\alpha \succ \frac{\beta}{\alpha} z_\alpha + \frac{\alpha - \beta}{\alpha} a_n = z_\beta. \end{aligned}$$

If  $\alpha \geq \beta$ , then either  $z_\alpha = z_\beta$  or  $z_\alpha \succ z_\beta$ . Thus (25) holds.

For all  $x \in X_A$ , let

$$\begin{aligned} u(x) &= \inf\{\alpha \in [0, 1] : z_\alpha \succeq x\} \\ v(x) &= \sup\{\beta \in [0, 1] : x \succeq z_\beta\}. \end{aligned} \quad (26)$$

By (23), these values are well-defined. Show that  $u = v$ . Indeed,  $z_\alpha \succeq x \succeq z_\beta$  implies  $\alpha \geq \beta$  and hence,  $u(x) \geq v(x)$ . Suppose that  $u(x) > v(x)$ . Then there is  $\gamma$  such that

$u(x) > \gamma > v(x)$ . By (26),  $u(x) > \gamma$  implies  $x \succ z_\gamma$ . By (26),  $\gamma > v(x)$  implies  $z_\gamma \succ x$ . By contradiction,  $u(x) = v(x)$ .

By (26),  $u$  is not constant because  $u(a_1) = 1$  and  $u(a_n) = 0$ . Show that  $u$  is mixture linear. By Independence, for any  $x, y \in X$  and  $\alpha, \beta, \gamma \in [0, 1]$  such that  $z_\alpha \succeq x$  and  $z_\beta \succeq y$ ,

$$z_{\gamma\alpha + (1-\gamma)\beta} = \gamma z_\alpha + (1-\gamma)z_\beta \succeq \gamma x + (1-\gamma)z_\beta \succeq \gamma x + (1-\gamma)y.$$

Thus  $u(\gamma x + (1-\gamma)y) \leq \gamma\alpha + (1-\gamma)\beta$ . By definition of  $u$ ,  $\alpha$  and  $\beta$  can be arbitrarily close to  $u(x)$  and  $u(y)$  respectively. Thus

$$u(\gamma x + (1-\gamma)y) \leq \gamma u(x) + (1-\gamma)u(y).$$

Analogously,  $v(\gamma x + (1-\gamma)y) \geq \gamma v(x) + (1-\gamma)v(y)$ . As  $u = v$ , then  $u \in \mathcal{U}_A$ .

For all  $x, y \in X_A$ , let  $\gamma = \frac{1}{2}u(x) + \frac{1}{2}u(y)$ . By (26),

$$u(x) > u(y) \quad \Rightarrow \quad u(x) > \gamma > v(y) \quad \Rightarrow \quad x \succ z_\gamma \succ y \quad \Rightarrow \quad x \succ y.$$

Thus representation (24) holds.

Suppose that (24) holds with another  $w \in \mathcal{U}_A$ . As  $a_1 \succeq a_n$ , then  $w(a_1) \geq w(a_n)$ . If  $w(a_1) = w(a_n)$ , then by (23),  $w(a_1) \geq w(z) \geq w(a_n)$  for all  $z \in X_A$ , and  $w$  is constant on  $X_A$ . By contradiction,  $w(a_1) > w(a_n)$ . For all  $x \in X_A$ ,

$$w'(x) = \frac{w(x) - w(a_n)}{w(a_1) - w(a_n)}.$$

Then  $w' \in \mathcal{U}_A$  is a positive linear transformation of  $w$  such that  $w'(a_1) = 1$  and  $w'(a_n) = 0$ . Representation (24) still holds for  $w'$ . As  $\alpha = w'(z_\alpha) \geq w'(x)$  for all  $\alpha$  such that  $z_\alpha \succeq x$ , then  $u(x) \geq w'(x)$ . As  $\beta = w'(z_\beta) \leq w'(x)$  for all  $\beta$  such that  $x \succeq z_\beta$ , then  $w'(x) \geq v(x)$ . Thus  $w'(x) = u(x)$ , and  $w$  is a positive linear transformation of  $u$ .  $\square$

As  $\succeq$  is stable, then there exist  $x^*, y^* \in X$  that satisfy (6). Take any menu  $A = \{a_1, a_2, \dots, a_n\}$  such that  $x^*, y^* \in A$ , and

$$a_1 \succeq a_2 \succeq \dots \succeq a_n.$$

As  $x^* \succ y^*$ , then  $a_1 \succ a_n$ . By Lemma A.4, there is  $u \in \mathcal{U}_A$  such that (24) holds. By (24),  $u(x^*) \geq u(y^*)$ . Suppose that  $u(x^*) = u(y^*)$ . Then for all  $\gamma \in (0, 1]$ ,

$$u(\gamma y^* + (1-\gamma)a_1) > u(\gamma x^* + (1-\gamma)a_n)$$

and hence,  $\gamma y^* + (1 - \gamma)a_1 \succ \gamma x^* + (1 - \gamma)a_n$ . Thus  $y^* \succeq^* x^*$ . This contradicts (6). Thus,  $u(x^*) > u(y^*)$ .

By Lemma A.4, there is a unique  $u_A \in \mathcal{U}_A$  that satisfies (24), and

$$u_A(x^*) = 1 > 0 = u_A(y^*).$$

For each  $x \in X$ , define

$$u(x) = u_{\{x^*, y^*, x\}}(x).$$

Take any  $x, y \in X$  and  $\gamma \in [0, 1]$ . Let  $B = \{x^*, y^*, x, y, \gamma x + (1 - \gamma)y\}$ . The uniqueness of the functions  $u_B$  and  $u_{\{x^*, y^*, x\}}$  implies that  $u_B = u_{\{x^*, y^*, x\}}$  on  $X_{\{x^*, y^*, x\}}$ . Thus  $u(x) = u_B(x)$ . Similarly,  $u(y) = u_B(y)$  and

$$u(\gamma x + (1 - \gamma)y) = u_B(\gamma x + (1 - \gamma)y).$$

As  $u_B$  satisfies (24) on  $X_B$ , then

$$x \succeq y \quad \Rightarrow \quad u(x) = u_B(x) \geq u_B(y) = u(y) \quad \Rightarrow \quad u(x) \geq u(y).$$

As  $u_B$  is mixture linear, then

$$u(\gamma x + (1 - \gamma)y) = u_B(\gamma x + (1 - \gamma)y) = \gamma u_B(x) + (1 - \gamma)u_B(y) = \gamma u(x) + (1 - \gamma)u(y)$$

and hence,  $u$  is mixture linear as well. Thus  $u \in \mathcal{U}$  and (24) holds.

Suppose that  $L$  is finite dimensional. Then there is a menu  $A$  such that the affine hull of  $A$  equals the affine hull of  $X$ . Index  $A = \{a_1, \dots, a_n\}$  so that

$$a_1 \succeq a_2 \succeq \dots \succeq a_n,$$

and let  $a_0 = \sum_{i=1}^n \frac{1}{n} a_i$ . As  $X$  belongs to the affine hull of  $A$ , then any  $x \in X$  can be written as an affine combination

$$x = \sum_{i=1}^n \gamma_i a_i \quad \text{such that} \quad \sum_{i=1}^n \gamma_i = 1.$$

Thus for any  $x \in X$ , there is  $\varepsilon > 0$  such that

$$0 \leq \gamma \leq \varepsilon \quad \Rightarrow \quad \gamma x + (1 - \gamma)a_0 \in X_A. \quad (27)$$

As  $\theta$  is non-null, then there are  $x, y \in X$  such that  $x \succ y$ . Take  $\varepsilon > 0$  such that  $X_A$  contains both  $\varepsilon x + (1 - \varepsilon)a_0$  and  $\varepsilon y + (1 - \varepsilon)a_0$ . By Independence and (23),

$$x \succ y \quad \Rightarrow \quad \varepsilon x + (1 - \varepsilon)a_0 \succ \varepsilon y + (1 - \varepsilon)a_0 \quad \Rightarrow \quad a_1 \succ a_n.$$

By Lemma A.4, there is  $u \in \mathcal{U}_A$  that satisfies (24) on  $X_A$ .

Take  $x^* = \frac{a_0 + a_1}{2}$  and  $y^* = \frac{a_0 + a_n}{2}$ . Take any  $z_x, z_y \in X$ . By (27), there is  $\varepsilon > 0$  such that  $X_A$  contains both  $\varepsilon z_x + (1 - \varepsilon)x^*$  and  $\varepsilon z_y + (1 - \varepsilon)y^*$ . As  $u$  is non-constant, then by (23),  $u(a_1) > u(a_n)$ . As  $u$  is mixture linear, then  $u(x_*) > u(y^*)$  and hence,

$$u(\varepsilon z_x + (1 - \varepsilon)x^*) > u(\varepsilon z_y + (1 - \varepsilon)y^*)$$

for sufficiently small  $\varepsilon$ . By (24),  $\varepsilon z_x + (1 - \varepsilon)x^* \succ \varepsilon z_y + (1 - \varepsilon)y^*$ . As  $z_x, z_y \in X$  are arbitrary, then  $\theta$  is stable.

### A.3 Proofs of Corollary 6

Let  $X = \Delta^Q$  be the set of all lotteries on some interval  $Q \subset \mathbb{R}$  of monetary prizes. Suppose that  $\rho_\mu$  satisfies ARA. Take any type  $\theta \in \Theta_\mu$ . Fix any payoffs  $q, r \in Q$  such that  $q > r$ . By ARA,  $q\theta^*r$  because  $\rho_\mu(r, \{q, r\}) = 0$ . Fix any lotteries  $z_q, z_r \in X$ . By ARA, there is  $\varepsilon > 0$  such that

$$\varepsilon z_q + (1 - \varepsilon)q\theta^*\frac{q+r}{2}.$$

By Independence, for all  $\alpha \in [0, \varepsilon]$ ,  $\alpha z_q + (1 - \alpha)q\theta^*\frac{q+r}{2}$  is preserved. Take  $\alpha$  sufficiently small so that

$$E(\alpha z_r + (1 - \alpha)r) = \alpha E(r) + (1 - \alpha)r < \frac{q+r}{2}.$$

Then by ARA,

$$\alpha z_q + (1 - \alpha)q\theta^*\frac{q+r}{2}\theta^*\alpha z_r + (1 - \alpha)r.$$

Thus the type  $\theta$  satisfies stability for  $x^* = q$  and  $y^* = r$ . By Theorem 1,  $\theta$  maximizes some  $u \in \mathcal{U}$ . As  $u$  is non-constant and has the expected utility form, then  $u$  is non-constant on  $Q$ . By Theorem 1, there are two possible cases. First, if  $\theta$  is stable

### A.4 Proof of Corollary 7

### A.5 Proof of Theorem 8

Suppose that  $\succeq$  is represented by (16) for some  $u \in \mathcal{U}$  and type  $\psi$  in  $L_u$ . Completeness is obvious. To show transitivity, take any  $x, y, z \in X$  such that  $x \succeq y \succeq z$ . Then  $u(x) \geq$

$u(y) \geq u(z)$ . If  $u(x) > u(z)$ , then  $x \succeq z$  by (16). If  $u(x) = u(z)$ , then  $x \psi y \psi z$  and hence,  $x \psi z$ . By (16),  $x \succeq z$ . Next,  $\theta$  satisfies Independence because  $u$  is mixture linear, and the tie-breaker  $\psi$  obeys Independence. Stability has been already derived from representation (5) and a fortiori, from (16).

Representation (16) in Theorem 8 implies that for any stable type  $\theta$ ,

$$\theta \text{ is total in } X \iff \psi \text{ is total in } L_u.$$

Indeed, if  $a \sim_\psi b$  for some distinct elements  $a, b \in L_u$ , then by Independence,  $\frac{a-b}{2} \sim_\psi 0$ . Take  $x, y \in X$  and  $\alpha \geq 0$  such that  $\alpha(x - y) = \frac{a-b}{2} \neq 0$  and  $u(x) = u(y)$ . By Independence,  $x - y \sim_\psi 0$ . By (16),  $x \sim_\theta y$ , and  $\theta$  is not total.

## B Select Slides for Experiment

### Certainty as a Decoy: an Experiment to Test Menu Dependence in Heterogeneous Risk Attitudes

Kevin Peng

Department of Economics, University of California, Irvine

November 2, 2025

## Research Question and Contribution

On the aggregate level, *the distribution of risk attitudes can vary depending on what options are presented to the decision makers.*

- *Harrison, et al., 12*: individuals are risk neutral when lotteries are presented in a binary format  $L = \begin{pmatrix} \$x & \text{w.p. } \pi \\ \$y & \text{w.p. } 1 - \pi \end{pmatrix}$ .
- *Hermanns & Kokot, 23*: framing can affect levels of risk aversion.

**Question:** does the presence of a certain option in the menu make agents on average more risk averse?

- Most literature focuses on **framing** effects (*Crosta, et al., 23*).
- *Chen, et al., 24*: introducing a high-risk-high-reward lottery does not decrease risk aversion.

→ I designed the menus so a no-risk-low-reward is introduced.

→ **Risk aversion is observed to increase.**

## Monotonicity

When additional options are added to a menu, the probability of choosing a **pre-existing** option should **not increase**: for all  $x \in A$ ,  $A$ ,  $y$ ,

$$\rho(x, A) \geq \rho(x, A \cup \{y\})$$

- A stochastic choice rule that has a random utility representation is **monotone**.
- However, if agents become more risk averse with the expansion of a menu, *they may switch to a pre-existing option that is of lower risk.*
- **Goal:** violations of monotonicity in some observations.

⇒ a menu-dependent version of random utility is needed to model the choice data.

# Hypotheses

Introduce a no-risk-low-reward (certain) option:

$$A \equiv \{\text{low-risk } (l), \text{high-risk } (h)\} \rightarrow \{l, h, \text{certain } (c)\} \equiv A'.$$

**H1:** the proportion of agents that choose  $h$  will decrease.

**H2:** the proportion of agents that choose  $l$  will increase.

**H3:** the relative proportion of agents that chose  $l$  will increase over  $h$ :

$$\rho(l, A) \leq \frac{\rho(l, A')}{\rho(l, A') + \rho(h, A')}.$$

Note that  $H2 \implies H3 \implies H1$ .

→ **Decoy effects** and **extremeness aversion** suggest at least  $H1$ .

**H4:** exposure to  $c$  has an extended effect, even after it is removed.

→ *Hedgcock, Rao & Chen, 16*: decoy effects can last beyond removal.

→ *Pettibone & Wedell, 07*: phantom decoys (not selectable).

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## Menu Without a Certain Option

### Main Task 2

Choose one of the following options:



Click "Continue" to go on to the next task.

Continue

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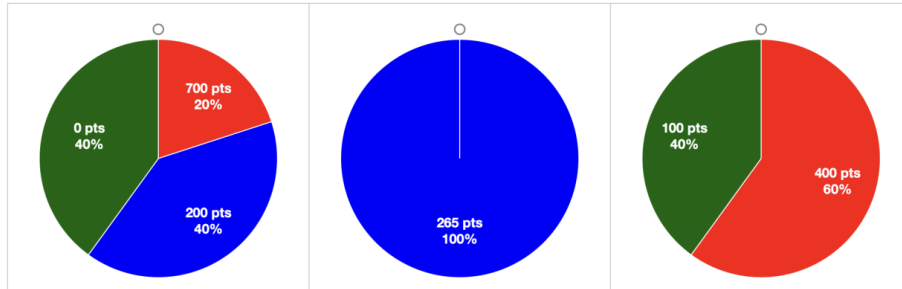
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# Menu With a Certain Option

## Main Task 42

Choose one of the following options:



Click "Continue" to go on to the next task.

Continue

## Main Tasks and Treatments

### Menu A

#### Main Task 2

Choose one of the following options:



Click "Continue" to go on to the next task.

Continue

### Menu A'

#### Main Task 42

Choose one of the following options:



Click "Continue" to go on to the next task.

Continue

- Each Menu  $A$  has a corresponding Menu  $A' \supseteq A$ .
- There are 26 pairs, with 52 total main tasks.

**T1:** Subjects face the 26  $A$  menus first, and then move on to 26  $A'$  menus.

**T2:** Subjects face the 26  $A'$  menus first, and then move on to 26  $A$  menus.

**T3:** Subjects face alternating menus  $A \rightarrow A' \rightarrow A \rightarrow \dots$ .

## Pilot Study

23 test subjects (UCI economics graduate students).

48 choices each, with a total of 1104 observations.

| (#)      | high-risk ( $h$ ) option |           | low-risk ( $l$ ) option |           | certain option |
|----------|--------------------------|-----------|-------------------------|-----------|----------------|
|          | no $c$                   | with $c$  | no $c$                  | with $c$  |                |
| $T1$ (8) | 95 (49%)                 | 77 (40%)  | 97 (51%)                | 84 (43%)  | 31 (16%)       |
| $T2$ (7) | 77 (46%)                 | 58 (35%)  | 91 (54%)                | 102 (61%) | 8 (5%)         |
| $T3$ (8) | 72 (38%)                 | 70 (36%)  | 120 (62%)               | 92 (48%)  | 30 (16%)       |
| Pool     | 244 (44%)                | 205 (37%) | 308 (56%)               | 278 (50%) | 69 (13%)       |
| Cond.    | 44%                      | 43%       | 56%                     | 57%       |                |

- $H1$  (decrease in high-risk choice) holds here.

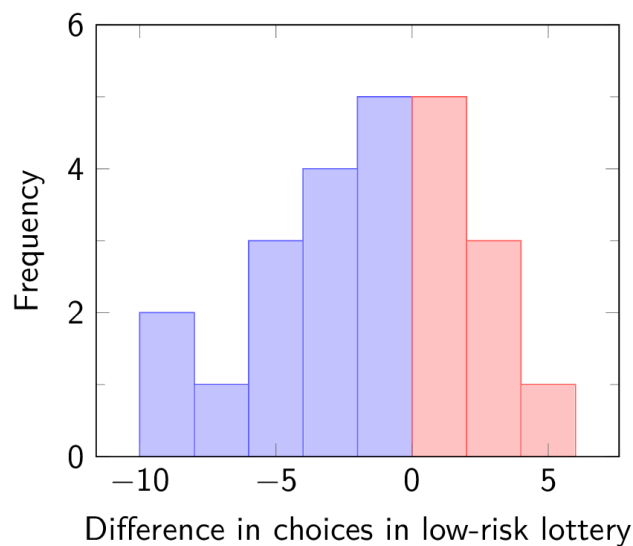
→ On average, agents become more risk averse with the presence of  $c$ .

- $H3$  does not seem statistically significant. ( $H2$  does not hold.)

→ Some people switch from  $l$  to  $c$ .

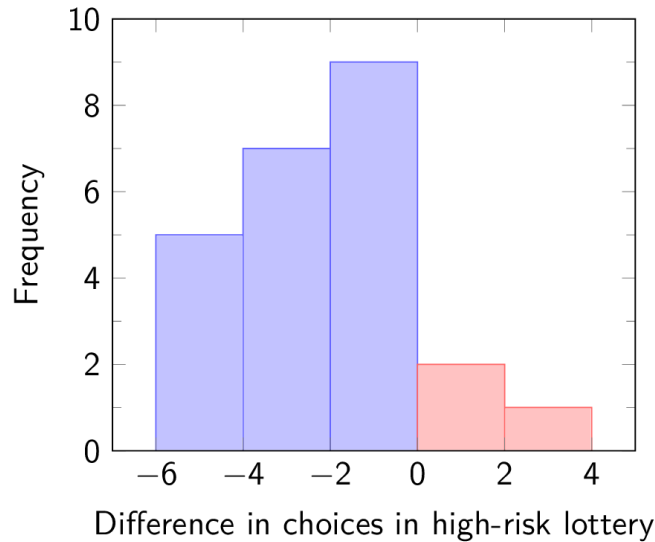
## Monotonicity: Low-Risk Option

When separately discussing each menu pair, monotonicity is violated in 9 of the 24 pairs when comparing the number of people that chose the low-risk option with or without the certain option ( $c$ ).



## Monotonicity: High-Risk Option

Monotonicity is violated in 3 of the 24 pairs when comparing the number of people that chose the *high-risk* option with or without the certain option.



## Logistic Regression with Random Effects

$$\log \left( \frac{P(Y_{ij}=1)}{1-P(Y_{ij}=1)} \right) = \beta_0 + \beta_1 \text{threeHigh}_{ij} + \beta_2 \text{threeLow}_{ij} + \beta_3 \text{switch}_{ij} + \beta_4 \text{certain}_{ij} + b_i$$

- $Y_{ij}$ : whether subject  $i$  chooses the riskier lottery in task  $j$ .
- $\text{threeHigh}_{ij}$ : whether the high-risk lottery in the task has three states.
- $\text{threeLow}_{ij}$ : whether the low-risk lottery in the task has three states.
- $\text{switch}_{ij}$ : the level of risk aversion (category) that would switch from high- to low-risk.
- $\text{certain}_{ij}$ : whether a certain lottery is present in the task.
- $b_i \sim \mathcal{N}(0, \sigma_b^2)$ : random intercept for subject  $i$ .

## Estimation

|             | Estimate | Std. Error | z value | Pr(>  z )  |
|-------------|----------|------------|---------|------------|
| (Intercept) | -3.562   | 0.319      | -11.150 | < 2e-16*** |
| threeHigh   | -0.165   | 0.140      | -1.174  | 0.240      |
| threeLow    | 0.049    | 0.140      | 0.353   | 0.724      |
| switch      | 0.603    | 0.046      | 13.155  | < 2e-16*** |
| certain     | -0.382   | 0.141      | -2.719  | 0.007**    |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**Table 1:** Fixed effect estimates from mixed-effects logistic regression predicting whether a subject will choose the high-risk lottery.

## Persistent Effect of Exposure ( $H4$ )

There should be higher level of risk aversion in the  $A$  portion (no  $c$ ) for subjects faced with  $A'$  (with  $c$ ) first ( $T2$ ) than those with  $A$  first ( $T1$ ).

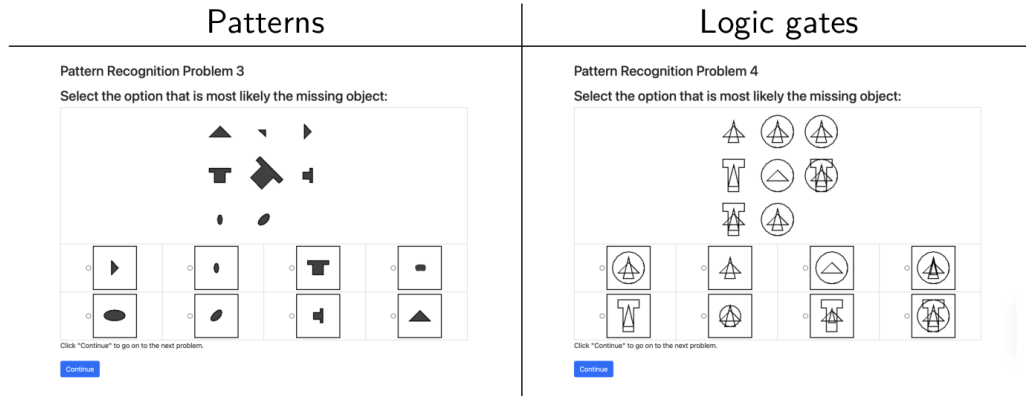
|      | high-risk option  |          | low-risk option |           | certain option |
|------|-------------------|----------|-----------------|-----------|----------------|
|      | no $c$            | with $c$ | no $c$          | with $c$  |                |
| $T1$ | 95 ( <b>49%</b> ) | 77 (40%) | 97 (51%)        | 84 (43%)  | 31 (16%)       |
| $T2$ | 77 ( <b>46%</b> ) | 58 (35%) | 91 (54%)        | 102 (61%) | 8 (5%)         |

- $T3$ 's level of risk aversion should increase over time. The starting level should be similar to  $T1$  and end similar to  $T2$ .

→ Plan: compare first 10 tasks with last 10 tasks.

# Cognitive Test: Pattern Recognition

Matzen, et al., 10: a non-branded version of Raven's Progressive Matrices.



- 8 problems with progressive difficulty.
- Plan: find relationship between cognition and menu-dependent risk attitudes. Cognizant agents less likely to be affected by the decoy?

## Conclusion

Findings in the pilot study:

- Agents are on average **more risk averse** when a certain option is presented.
- *Monotonicity* is **violated** in a third of the scenarios.

To do for the main study (around 270 subjects, to be collected in ESSL):

- Observe what types of menus are more likely to violate monotonicity.
- Analysis on persistence of decoy effects (*H4*).
- Relate cognition to sensitivity to decoy effects.

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